

Guide to Becoming a Data Analyst: Statistics & Mathematics

This guide provides a detailed discussion of 11 essential Statistics and Mathematics topics required for Data Analytics.

Descriptive Statistics (বর্ণনামূলক পরিসংখ্যান)

1. Definition

What is it? Descriptive Statistics is a method of collecting, analyzing, and presenting data in such a way that the main characteristics of the data are easily understood. Essentially, it creates a summary of the data we have. It does not make predictions but provides a clear picture of “what is happening currently” or “what happened in the past.”

Explanation in simple terms with a daily life example: Suppose you have the math scores of 50 students in a class. If someone asks you, “How are the students performing in math?” you would not list all 50 scores individually. Instead, you might say, “The average score is 75, the highest score is 95, and the lowest score is 40.” By representing the entire dataset with just these three numbers, you are using Descriptive Statistics.

1. Why is it important?

Why is it used in Data Analytics?

- **Understanding Data:** It is impossible to make sense of large amounts of raw data. Descriptive Statistics transforms that data into meaningful information.
- **Pattern Recognition:** Helps identify specific trends or patterns within the data.
- **Outlier Detection:** Assists in finding abnormal values in the data (e.g., if everyone's salary is between 20,000-30,000 but one person's salary is 500,000).

Importance in Business Decisions:

- **Performance Evaluation:** Knowing total sales last month, average sales, etc., helps understand the current state of the business.
- **Quick Decision Making:** Management can make fast decisions through dashboards or reports. For example, knowing which product sells the most (Mode) helps increase stock for that product.

1. How do companies use it?

Real-world examples:

- **E-commerce (e.g., Daraz):** To calculate customers' Average Order Value and identify the most sold products (Mode).

- Banking (e.g., bKash): To understand the average daily transaction amount and the spread of transactions (Standard Deviation).
- Healthcare (e.g., Square Hospital): To analyze the average number of daily patient admissions and the average age of patients.
- Telecom (e.g., Grameenphone): To calculate average call duration and data usage.
- Marketing (e.g., Ogilvy): To measure average clicks on an advertisement (Click-through Rate) and identify which age group watches it the most.
- Retail (e.g., Shwapno): To analyze which day of the week has the highest customer footfall and average spending.
- Manufacturing (e.g., Walton): To determine the average number of refrigerators produced daily and the defect rate.

1. How does a Data Analyst use it?

How to use it with Excel?

- Mean: =AVERAGE(A2:A100)
- Median: =MEDIAN(A2:A100)
- Mode: =MODE(A2:A100)
- Standard Deviation: =STDEV.S(A2:A100) (for sample)

Using Data Analysis Toolpak: Go to the Data tab in Excel, select Data Analysis > Descriptive Statistics to generate a complete summary with one click.

How to use it with SQL?

```
SELECT
    AVG(sales_amount) AS mean_sales,
    MIN(sales_amount) AS min_sales,
    MAX(sales_amount) AS max_sales,
    COUNT(order_id) AS total_orders,
    SUM(sales_amount) AS total_revenue
FROM sales_data;
```

How to use it with Python?

```
import pandas as pd

# Load data
df = pd.read_csv('sales_data.csv')

# Generate Descriptive Statistics
summary = df['sales_amount'].describe()
print(summary)

# Calculate individually
mean_val = df['sales_amount'].mean()
median_val = df['sales_amount'].median()
mode_val = df['sales_amount'].mode()[0]
std_dev = df['sales_amount'].std()
```

How to use it in Power BI?

Create Measures using DAX (Data Analysis Expressions)

Done:

```
Total Sales: Total Sales = SUM(Sales[Amount])

Average Sales: Avg Sales = AVERAGE(Sales[Amount])

Median Sales: Median Sales = MEDIAN(Sales[Amount])
```

1. Real Business Scenario

A real problem of a company: Suppose Foodpanda noticed that deliveries in a specific area are often delayed. Customers are complaining. Management wants to know the actual status of delivery times.

How it is solved (Step by step):

1. Data Collection: Data of all delivery times (in minutes) from the last 1 month in that area was collected.
2. Mean Calculation: It was found that the average delivery time is 45 minutes (which is higher than normal).
3. Median Calculation: But the median was found to be 35 minutes.

4. Analysis: The difference between Mean and Median indicates that there are outliers in the data. That is, most deliveries are happening in 35 minutes, but some deliveries might have taken 2-3 hours (due to rain or traffic), which caused the average (Mean) to rise to 45 minutes.
5. Standard Deviation: The standard deviation was found to be 15 minutes. This means there is a lot of variation in delivery times.
6. Conclusion: Management understood that not all riders are delayed. In some specific cases, there are significant delays. They then identified the causes of those outliers (such as delays at specific restaurants or traffic jams on certain roads) and resolved them.

2. Interview Perspective

Question 1: What is the difference between Mean and Median? When would you use which?

Sample Answer: Mean is the mathematical average of all values, while Median is the middle value when data is arranged in order. If there are outliers (extremely high or low values), Mean is affected but Median is not. Therefore, when outliers exist, it is better to use Median, for example, when calculating people's income or salary.

Question 2: What do you understand by Standard Deviation?

Sample Answer: Standard Deviation indicates how far or spread out the data points are from their mean. If Standard Deviation is low, data points are close to the mean (consistent); if high, data points are widely spread (volatile).

Question 3: What is the relationship between Variance and Standard Deviation?

Sample Answer: Variance is the average of the squared distances of each data point from the mean. The square root of Variance gives the Standard Deviation. Standard Deviation has the same unit as the original data, making it easier to interpret.

Question 4: What is an Outlier and how does it affect Descriptive Statistics?

Sample Answer: An Outlier is a value in the dataset that is abnormally high or low compared to others. It significantly affects the Mean and Variance, which can lead to a distorted picture of the data.

Question 5: What is Skewness?

Sample Answer: Skewness is the asymmetry of the data distribution. If the tail is longer on the right side, it is called Positive Skewness ($\text{Mean} > \text{Median}$). If the tail is longer on the left side, it is Negative Skewness ($\text{Mean} < \text{Median}$).

7. Beginner Mistakes

Common mistakes made by beginners:

1. Using Mean without checking for Outliers: Making wrong decisions by using Mean despite the presence of outliers.

2. Ignoring Standard Deviation: Making decisions based only on the mean without considering data spread or risk.
3. Confusing Correlation with Causation: Assuming that because two variables are correlated, one causes the other.
4. Avoiding Missing Data: Calculating mean or other statistics without properly handling missing values (nulls).
5. Using wrong charts: Using inappropriate visualizations without understanding data type (e.g., using Bar charts for time series instead of Line charts).

How to avoid these?

Always **create** a Boxplot **or** Histogram before data analysis **to check** distribution.

Always look **at** Median **and** Standard Deviation alongside Mean.

Make logical decisions **to** fill missing **values with** Mean/Median **or** exclude

1. Formula

2. Mean (Average - μ or $\bar{x}$):

$$\bar{x} = (\sum x_i) / n$$

$\sum x_i$ = sum of all data points

n = total number of data points

3. Median: After arranging data from smallest to largest:

If n is odd: $(n+1)/2$ th term

If n is even: Average of the $(n/2)$ th and $(n/2 + 1)$ th terms

1. Variance (ভেদাঙ্ক - σ^2 or s^2):

Population Variance: $\sigma^2 = \sum (x_i - \mu)^2 / N$

Sample Variance: $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$

Where:

x_i = each data point

$\bar{x}$ or μ = mean

N or n = total number of data points

1. Standard Deviation (পরিমিত ব্যবধান - σ or s):

$$\sigma = \sqrt{\text{Variance}}$$

2. Practical Exercise

Beginner Exercise (Simple):

3. Calculate the Mean: The ages of 5 friends are 22, 24, 23, 25, 26 respectively. What is their average age?
4. Calculate the Median: Salaries (in thousands) of 7 employees are: 15, 18, 20, 22, 25, 30, 100. What is the median salary?
(Note that 100 is an outlier.)
5. Calculate the Mode: Shoe sizes sold in a store are: 7, 8, 8, 9, 8, 10, 7. Which shoe size was sold the most?

Intermediate Exercise (Moderate difficulty):

1. Standard Deviation: Sales of a company over the last 5 months (in lakhs): 10, 12, 11, 13, 14. Calculate the Variance and Standard Deviation.
2. Missing Value Handling: Ratings of 10 customers are given, but 2 ratings are missing: 4, 5, 3, 4, Null, 5, 2, 4, Null, 5. Fill the missing values with the mean and then calculate the new mean.

Real Business Exercise (Real company problem):

Problem: Pathao ride-sharing service has provided fare data for various trips in Dhaka city over the past week (in Taka): 120, 150, 130, 200, 180, 250, 1500 (Outlier), 140, 160, 170. Tasks:

- Calculate the Mean and Median fare.
- Explain the effect of the outlier.
- Which metric (Mean or Median) would you report to the company management and why?

1. Career Perspective

- Junior Data Analyst: Primarily uses Excel or SQL to clean data and calculate Mean, Median, Mode to create basic reports or dashboards.
- Data Analyst: Uses Python/R and Power BI/Tableau to analyze data distribution, outliers, and standard deviation. Tries to find the “why” behind the data.
- Senior Data Analyst: Uses insights from descriptive statistics to help the company make strategic decisions. Prepares data for predictive modeling.

- **Business Analyst:** Focuses more on the business impact rather than the technical statistical aspects. For example, “What impact will the decrease in average sales have on company revenue and how can it be increased?”

Summary and Cheat Sheet

Key points summarized:

- **Descriptive Statistics** summarize data and highlight its main characteristics.
- **Measures of Central Tendency:** Mean, Median, Mode — these indicate the center of the data.
- **Measures of Dispersion:** Variance, Standard Deviation, Range — these indicate how spread out the data is.
- When outliers are present, Median is more reliable than Mean.

Quick Reference Cheat Sheet:

Metric	When to Use?	Excel Formula
Mean	When data is normally distributed and there are no outliers.	=AVERAGE()
Median	When data has outliers or is skewed (e.g., income, age).	=MEDIAN()
Mode	For categorical data (e.g., most popular color or size).	=MODE()
Standard Deviation	To understand data variation or risk.	=STDEV.S()
Range	To understand the difference between max and min values.	=MAX() - MIN()

Probability (সম্ভাব্যতা)

1. Definition

What is it? Probability is the mathematical measure of the likelihood or chance of an event occurring. It is expressed as a number between 0 and 1, where 0 means the event is impossible (will not happen) and 1 means the event is certain (will definitely happen).

Explanation in Simple Bengali and Examples from Daily Life: In simple terms, the chance of something happening is called Probability. For example, if you toss a coin, what is the probability of getting 'Head'? Since a coin has two sides (Head and Tail), the probability of getting Head is 1 out of 2, i.e., $\frac{1}{2}$ or 0.5 (50%). Another example is weather forecasting. When it is said "There is a 70% chance of rain today," it means that based on past data and current weather conditions, it has been observed that it rains on 70 out of 100 such days.

1. Why is it Important?

Why is it used in Data Analytics? In data analytics, we analyze past data to predict the future. Since the future is always uncertain, Probability helps us measure this uncertainty mathematically. It is the foundation for building Machine Learning models, Risk Assessment, and Predictive Modeling.

What is its importance in Business Decision-making? Every business decision involves risk. By using Probability, companies can understand which decisions carry less risk and have a higher chance of profit. For example, if a new product is launched, what is the probability of its success? Or if a customer takes a loan, what is the probability that they will default? This helps in making the right business decisions.

2. How Do Companies Use It?

- E-commerce (e.g., Daraz, Amazon): To determine the probability that a customer will buy a product after viewing it (Conversion Probability). Based on this, discounts are offered.
- Banking (e.g., bKash, City Bank): In Credit Risk Analysis. The probability that a customer will default on a loan is calculated before approving the loan.
- Healthcare (e.g., Square Hospital): To analyze patient symptoms and test reports to determine the probability of a specific disease.
- Telecom (e.g., Grameenphone): For Customer Churn Prediction. The probability that a customer will switch to another operator next month is calculated to offer special deals.
- Marketing (e.g., Facebook Ads): To calculate the probability that a user will click on an advertisement (Click-Through Rate Probability) and show ads accordingly.
- Retail (e.g., Shwapno): In Inventory Management. To estimate the probability of demand for a specific product next week and stock accordingly.
- Manufacturing (e.g., Walton): In Quality Control. To determine the probability that a product coming off the production line is defective.

1. How Does a Data Analyst Use It?

How to use it with Excel? Excel has various functions to calculate Probability.

Functions/Formulas: PROB(), NORM.DIST(), BINOM.DIST()

Example: If you have data of 50 customers and want to find the probability that a customer

is older than 30, you can use COUNTIF(range, ">30")/COUNT(range) to calculate a simple probability.

How to use it with SQL? Although SQL does not have a direct Probability function, we can calculate it using Aggregate Functions.

Example Query:

```
SELECT
    COUNT(CASE WHEN purchase_status = 'Bought' THEN 1 END) * 1.0 / COUNT
FROM customer_data;
```

This query calculates the probability of customers who have bought a product out of total customers.

How to use it with Python? In Python, libraries like scipy.stats and numpy are used for Probability calculations.

Code Example:

```
import numpy as np
# Simulation of tossing a coin 100 times (0=Tail, 1=Head)
tosses = np.random.choice([0, 1], size=100)
probability_of_head = np.sum(tosses) / len(tosses)
print(f"Probability of getting Head: {probability_of_head}")
```

How to use it with Power BI? Probability can be calculated in Power BI using DAX (Data Analysis Expressions).

DAX/Measure Example:

Churn Probability =

DIVIDE(

```
CALCULATE(COUNT(Customers[CustomerID]), Customers[Status] = "Churned")
COUNT(Customers[CustomerID]),
0
```

)

This measure shows the probability of customers who have churned out of the total customers.

1. Real Business Scenario

A real problem faced by a company: Suppose “Grameenphone” is observing that many of their prepaid customers are switching to other operators (Customer Churn). They want to identify which customers are likely to churn...

Most likely to leave, so that they can be retained with special offers.

How it is solved (Step by step):

1. Data Collection: Customer data from the last 6 months was collected (call duration, recharge amount, internet usage, number of calls to customer care).
 2. Probability Calculation: It was found that among those who recharge less than 100 Taka per month and have called customer care more than 3 times, 80% of customers switched to another operator.
 3. Applying Probability: Now, analyzing current customer data, those who have these characteristics were identified. Their Churn Probability is 0.80 (or 80%).
 4. Decision Making: For those with a Churn Probability above 70%, Grameenphone targeted them with free internet or special call rate offers.
 5. Result: As a result, the customer churn rate significantly decreased.
- #### 6. Interview Perspective
7. What is Probability and why is it important? Sample Answer: Probability is the mathematical likelihood of an event occurring, ranging between 0 and 1. It is important in data analytics because it helps measure uncertainty and make future predictions based on data.
 8. What are Mutually Exclusive Events? Sample Answer: When two events cannot happen at the same time, they are called Mutually Exclusive Events. For example, when tossing a coin, it cannot land on both Head and Tail simultaneously.
 9. What is the difference between Independent and Dependent Events? Sample Answer: In an Independent Event, the outcome of one event does not depend on the other (e.g., tossing a coin twice). In a Dependent Event, the outcome of one event affects the other (e.g., drawing a card from a packet without replacement and then drawing another card).
 10. What is Conditional Probability? Sample Answer: Conditional Probability is the probability of an event occurring given that another event has already occurred. For example, if a customer has bought a laptop, what is the probability that they will buy a mouse?

11. How does Probability work in Normal Distribution? Sample Answer: In Normal Distribution, data forms a bell-shaped curve. According to the Empirical Rule, 68% of data lies within one Standard Deviation from the Mean, which helps us understand the spread and likelihood within the data.
12. Beginner Mistakes
What mistakes do beginners make?
13. Probability greater than 1: Many mistakenly calculate Probability values greater than 1 or less than 0, which is impossible.
14. Confusing Independent and Dependent Events: Adding or multiplying probabilities without understanding whether events are independent or dependent.
15. Confusing Correlation with Causation: Assuming that because two events are related, one causes the other.
16. Base Rate Fallacy: Ignoring the overall probability (base rate) and making decisions based only on specific information.
17. Small Sample Size: Making significant decisions based on Probability calculated from very limited data.

How to avoid these?

- Always remember that Probability values must be between 0 and 1.
- Analyze carefully whether events are Independent or Dependent.
- Use large and appropriate sample sizes.
- Understand the context behind the data.

1. Formula

All important formulas:

Number of Favorable Outcomes

2. Basic Probability: $P(A) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$

Total Number of Possible Outcomes

Explanation: $P(A)$ is the probability of event A occurring. It is calculated by dividing the number of favorable outcomes by the total number of possible outcomes.

3. Complementary Rule: $P(A') = 1 - P(A)$

Explanation: $P(A')$ is the probability that event A does not occur. If the probability of an event occurring is known, subtracting it from 1 gives the probability of the event not occurring.

4. Addition Rule (Mutually Exclusive): $P(A \text{ or } B) = P(A) + P(B)$

Explanation: If two events cannot occur simultaneously, the probability of either event occurring is the sum of their individual probabilities.

5. Multiplication Rule (Independent Events): $P(A \text{ and } B) = P(A) \times P(B)$

Explanation: The probability of two independent events occurring together is the product of their individual probabilities.

$P(A \text{ and } B)$

6. Conditional Probability: $P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$

$P(B)$

Explanation: $P(A|B)$ is the probability of event A occurring given that event B has already occurred.

Probability of an Event Occurring.

7. Practical Exercise Beginner Exercise (Simple):

1. There is a box containing 3 red balls and 2 blue balls. If you pick a ball with your eyes closed, what is the probability that it is red? (Data: Red=3, Blue=2, Total=5)
2. On an e-commerce website, out of 100 visitors, 5 purchase a product. What is the probability that a new visitor will purchase a product?
3. If a coin is tossed 3 times, what is the probability of getting at least one Head?

Intermediate Exercise (Moderately Difficult):

1. At a bank, it is observed that 40% of credit card users take personal loans. If 20% of the bank's customers use credit cards, what is the probability that a randomly selected customer is both a credit card user and a personal loan borrower?

2. In a factory, Machine A produces 60% of the products and Machine B produces 40%. Machine A produces 2% defective products, and Machine B produces 3% defective products. What is the probability that a randomly selected product is defective?

Real Business Exercise (Real Company Problem): Problem: "Daraz" observed that among customers who shop in the Electronics category, 30% also buy Accessories. Today, 500 customers shopped in the Electronics category. Task: Using the probability of buying Accessories, calculate how many customers are likely to buy Accessories and based on this information, how can Daraz increase their revenue? 10. Career Perspective

Junior Data Analyst: Typically uses basic probability to explore data. For
Data Analyst: Uses probability for A/B Testing and Hypothesis Testing. For
Senior Data Analyst: Builds complex predictive models. For example, using
Business Analyst: Analyzes probability results to develop strategies for .

Summary and Cheat Sheet Key points summarized:

Probability is a mathematical measure of uncertainty (value between 0 and 1).
It is used in data analytics to predict the future and reduce business risk.
Core concepts include Independent, Dependent, Mutually Exclusive, and Conditional.
It can be calculated using various functions and queries in Excel, SQL, Python, etc.

Quick Reference Cheat Sheet:

Probability Range: $0 \leq P(A) \leq 1$

Certain Event: $P(A) = 1$

Impossible Event: $P(A) = 0$

Not A (Complement): $P(A') = 1 - P(A)$

A or B (Mutually Exclusive): $P(A \cup B) = P(A) + P(B)$

A and B (Independent): $P(A \cap B) = P(A) \times P(B)$

Conditional (A given B): $P(A|B) = P(A \cap B) / P(B)$

Inferential Statistics ()

1. Definition What is it? Inferential Statistics is a branch of statistics that helps make conclusions or inferences about an entire population by analyzing data from a small sample. Explanation in simple terms and everyday example: Suppose you are cooking a pot of rice. To check if the rice is cooked, you don't taste every grain in the pot. Instead, you pick 2-3 grains (sample) and based on that, infer whether the entire pot of rice (population) is cooked. This is the core idea of Inferential Statistics.
2. Why is it important? Why is it used in Data Analytics? Collecting data from the entire population is often impossible, time-consuming, and expensive. Therefore, data analytics uses sample data to make accurate decisions about the whole population. What is its importance in Business Decision-making? It is not feasible for companies to know the opinions or behaviors of millions of customers. So, they conduct surveys or tests on a sample of customers and make decisions about the entire customer base, which helps in marketing, product development, and risk management.

3. How do companies use it? Real-world examples:

E-commerce (e.g., Daraz): Conducts A/B testing to see which website design or offer...

Attract more customers. By showing some customers the new design and analyzing their responses, decisions are made for the entire customer group.

Banking (e.g., bKash): By analyzing the transaction patterns of some customers, the credit risk or likelihood of loan default for the entire customer base is estimated.

Healthcare (e.g., Square Hospital): Clinical trials are conducted on some patients to verify the effectiveness of a new medicine, and based on that, a decision is made whether the medicine will work for everyone.

Telecom (e.g., Grameenphone): By analyzing call drop or network issue data from some customers, the overall network quality of the entire area is estimated.

Marketing (e.g., Unilever): A new product advertisement is shown to a focus group, and based on their feedback, it is predicted how the entire country's population will receive it.

Retail (e.g., Shwapno): By analyzing sales data from a few stores, future sales forecasting for all stores nationwide is estimated.

Manufacturing (e.g., Walton): Randomly selected products from the production line are tested to ensure quality control for the entire batch.

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1. How does a Data Analyst use these?

How to use Excel? Using the Data Analysis Toolpak in Excel, t-Test, ANOVA, and Regression can be performed.

Formula: =T.TEST(array1, array2, tails, type) – to check if there is a difference between the means of two samples.

How to use SQL? SQL is generally used for data extraction but basic statistics can also be derived.

```
SELECT
    AVG(sales_amount) as mean_sales,
    STDDEV(sales_amount) as std_dev_sales
FROM sales_data
WHERE region = 'Dhaka';
```

How to use Python? Python uses libraries like scipy.stats and statsmodels.

```
import scipy.stats as stats

# Independent t-test
group1 = [25, 30, 28, 35, 32]
group2 = [20, 22, 25, 28, 24]
t_stat, p_value = stats.ttest_ind(group1, group2)
print(f"P-value: {p_value}")
```

How to use Power BI? In Power BI, DAX is used for Confidence Intervals or Predictions.

```
Confidence_Interval_Upper =
AVERAGE(Sales[Amount]) + (1.96 * (STDEV.S(Sales[Amount]) / SQRT(COUNT(Sales[A
```

1. Real Business Scenario

A real company problem: Foodpanda wants to know if their new “Free Delivery” offer has actually increased the number of customer orders.

How this statistical concept is used to solve the problem (Step by step):

- Step 1: Formulate Hypotheses.
Null Hypothesis (H0): The new offer did not increase the number of orders.
Alternative Hypothesis (H1): The new offer increased the number of orders.

- Step 2: Collect Samples. 1000 customers were given the “Free Delivery” offer (Test Group) and 1000 were not (Control Group).
 - Step 3: Select Test. Since two separate groups are compared, an Independent Two-Sample t-Test will be used.
 - Step 4: Analyze P-value. Suppose after performing the t-Test, the P-value is found to be 0.02.
 - Step 5: Make Decision. Since $P\text{-value} (0.02) < 0.05$ (Significance Level), we reject the Null Hypothesis. That means the “Free Delivery” offer has indeed increased the number of orders.
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1. Interview Perspective

What types of interview questions can come from this topic?

2. What is the difference between Descriptive and Inferential Statistics?
3. What is a P-value and how do you interpret it?
4. Explain Confidence Interval in simple terms.
5. What is the Central Limit Theorem and why is it important?
6. How do you decide which statistical test (t-test, ANOVA, Chi-square) to use?

Sample Answers for each question:

1. Answer: Descriptive Statistics only summarize or describe data (e.g., Mean, Median). Inferential Statistics use sample data to make decisions or inferences about the entire population.
2. Answer: A P-value is a probability that indicates whether the observed result is due to chance or is actually true. If $P\text{-value} < 0.05$, we consider the result statistically significant.
3. Answer: A Confidence Interval is a range within which the true population parameter is likely to lie. For example, a 95% Confidence Interval means we are 95% confident that the true value falls within this range.

Within limits.

1. Answer: The Central Limit Theorem (CLT) states that regardless of the population distribution, if the sample size is large (generally $n > 30$), the distribution of the sample means will approximate a Normal Distribution. This forms the basis of Inferential Statistics.

2. Answer: The choice of test depends on the type of data and the number of groups. To compare the means of two groups, a t-test is used; for three or more groups, ANOVA is applied; and to examine relationships between categorical data, the Chi-square test is used.
3. Beginner Mistakes
What mistakes do beginners make?
4. Confusing Correlation with Causation (i.e., assuming that if two things happen together, one causes the other).
5. Misinterpreting the P-value (a P-value of 0.05 does not mean there is a 5% chance that H0 is true).
6. Taking a very small sample size, leading to inaccurate results.
7. Performing tests without checking for Outliers (abnormal data).
8. Using incorrect tests without checking the data distribution (e.g., applying parametric tests on non-normal data).

How to avoid these? Always visualize the data to check for Outliers and Distribution. Remember, Correlation does not imply Causation. Conduct Power Analysis to determine the correct Sample Size.

1. Formula
All important formulas:
2. Z-Score:
[$Z = \frac{X - \mu}{\sigma}$]
3. Confidence Interval (Mean):
[$CI = \bar{x} \pm Z \times \frac{s}{\sqrt{n}}$]
4. T-Statistic:
[$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$]

Explanation of each part/symbol in the formulas:

- (X) = Specific data point
- (μ) = Population Mean
- (σ) = Population Standard Deviation
- ($\bar{x}$) = Sample Mean
- (Z) = Z-value (according to Confidence Level)

- (s) = Sample Standard Deviation
- (n) = Sample Size

1. Practical Exercise

3 Beginner Exercises:

2. The average score of 10 students in a class is 75 with a Standard Deviation of 5. Calculate the 95% Confidence Interval.
3. Lifespans of bulbs from two different brands are given. Use a t-test to check if there is any difference between them. (Brand A: 1000, 1050, 980; Brand B: 1100, 1150, 1120)
4. A coin is tossed 20 times and lands heads 15 times. Is the coin biased? (Binomial Test)

2 Intermediate Exercises:

1. Conversion rates for three different designs (A, B, C) of an e-commerce site are given. Use ANOVA to find out which design is the best.
2. Conduct Regression Analysis to determine if there is any relationship between the age and income of people in a city.

1 Real Business Exercise:

Problem: Pathao has introduced a new incentive model for their riders. In the previous model, riders completed an average of 10 trips per day. Data from 50 riders under the new model shows they complete an average of 12 trips per day (Standard Deviation = 2). At a 95% Confidence Level, prove that the new model has indeed increased the number of trips.

1. Career Perspective

How does a Junior Data Analyst use this concept?

Juniors typically collect data for A/B tests under senior guidance and prepare reports by calculating basic t-tests or Confidence Intervals.

How does a Data Analyst use this?

They independently design hypotheses, select appropriate statistical tests, and extract insights from data using Python/R to provide to the business team.

How does a Senior Data Analyst use this?

They design complex experiments, perform Power Analysis to determine sample size, and help make strategic business decisions based on model results.

How does a Business Analyst use this?

They do not focus on the math behind statistics but use results (such as P-value, Confidence Interval) to explain to stakeholders which business strategies will work and which will not.

Summary and Cheat Sheet

Key points of the topic summarized: Inferential Statistics is the process of making decisions about a population based on sample data. Its foundation lies in Hypothesis Testing, Confidence Intervals, and P-values.

It is widely used in business for A/B testing, risk management, and forecasting. Quick Reference Cheat Sheet:

Test Name When to Use (কখন ব্যবহার করবেন) Example

1-Sample t- To compare the mean of one sample with a Is our delivery time less than 30 test specific value minutes?

2-Sample t- To compare the means of two different groups Is there a difference in average test shopping between men and women?

ANOVA To compare the means of 3 or more groups Which of the 3 different marketing

campaigns :

Chi-Square To examine the relationship between two Is there a relationship between Test categorical variables gender and product preference?

Regression To see the effect of one variable on another How much will sales increase if

advertising

Hypothesis Testing প্রকল্প পরীক্ষা ()

1. Definition (সংজ্ঞা) What is it? Hypothesis Testing or Project Testing is a method in statistics through which we verify whether an assumption or claim based on data or information is true or not. Simply put, it is the process of using data to find evidence for a concept. Explanation in simple Bengali and everyday life example: Suppose a tea shop owner in your area claims that on average 500 cups of tea are sold daily. This is a claim or assumption (Hypothesis). Now you want to verify whether this claim is true or not. Since you cannot collect data for a lifetime, you randomly collect data for 10 days. Based on this 10-day data, you mathematically try to prove whether the shop owner's claim is true or false. This entire process is Hypothesis Testing. There are two main parts:

Null Hypothesis (H_0): This is the default or general assumption. That is, "The shop owner's claim is correct, 500 cups of tea are sold daily."

Alternative Hypothesis (H1 or Ha): This is the opposite of the Null Hypothesis. That is, “The shop owner’s claim is not correct, 500 cups of tea are not sold daily.”

2. Why is it important? Why is it used in Data Analytics? In data analytics, we often work on samples because working with the entire population or complete data is time-consuming and costly. Hypothesis Testing helps us mathematically prove whether the results obtained from sample data apply to the entire population. It assists in making data-driven decisions rather than relying on assumptions. What is its importance in Business Decision-making? Making wrong decisions in business can cause losses worth millions. For example, a company launches a new marketing campaign. Hypothesis Testing is used to determine whether the campaign actually increased sales. It helps companies move from “I think” (Gut feeling) to “Data says” (Data-driven) decisions.

3. How do companies use it? In real life, Hypothesis Testing is widely used across various industries:

E-commerce (e.g., Daraz, Amazon): They use A/B Testing to change website design. For example, to check if changing the “Buy Now” button color from blue to red increases sales, Hypothesis Testing is used. Banking (e.g., bKash, City Bank): To test whether launching a new loan policy reduces the default rate (rate of loan non-payment). Healthcare (e.g., Square Hospital): To verify through clinical trials whether a new medicine or treatment method is more effective than the previous one. Telecom (e.g., Grameenphone): To test whether launching a new internet package increases customers’ data usage. Marketing (e.g., Unilever): Facebook ads vs. YouTube ads—Which one leads to more customer

To verify that the Acquisition Cost (Customer Acquisition Cost) is low.

Retail (e.g., Shwapno): To test whether changing the store layout or placing products on specific shelves increases sales.

Manufacturing (e.g., Walton): To verify whether producing with new machines reduces the product defect rate.

1. How does a Data Analyst use it?

How to use it with Excel?

Using the Data Analysis Toolpak in Excel, t-Test and z-Test can be easily performed.

Function/Formula: =T.TEST(array1, array2, tails, type)

Example: Using sales data from two different marketing campaigns (array1 and array2) to determine if there is a significant difference between them.

How to use it with SQL?

SQL is not directly designed for Hypothesis Testing, but it is essential for preparing data.

Example query:

```

SELECT
    campaign_type,
    AVG(sales_amount) as avg_sales,
    STDDEV(sales_amount) as std_dev,
    COUNT(*) as sample_size
FROM sales_data
GROUP BY campaign_type;

```

This data is then tested in Python or Excel.

How to use it with Python?

Using Python's scipy.stats library, Hypothesis Testing can be done very easily. Example code:

```

import scipy.stats as stats

# Sales data for two groups
group_a = [120, 130, 125, 140, 135]
group_b = [150, 160, 155, 165, 170]

# T-test
t_stat, p_value = stats.ttest_ind(group_a, group_b)

print(f"P-value: {p_value}")
if p_value < 0.05:
    print("Significant difference exists (Reject Null Hypothesis)")
else:
    print("No significant difference (Fail to reject Null Hypothesis)")

```

How to use it with Power BI?

Power BI does not have built-in features for Hypothesis Testing, but Z-score or T-score can be calculated using DAX, or Python/R scripts can be integrated.

DAX example (Z-score):

```

Z_Score =
VAR AvgSales = CALCULATE(AVERAGE(Sales[Amount]), ALL(Sales))
VAR StdDevSales = CALCULATE(STDEV.P(Sales[Amount]), ALL(Sales))
RETURN (SUM(Sales[Amount]) - AvgSales) / StdDevSales

```

1. Real Business Scenario

A real company problem: Suppose Foodpanda has developed a new routing algorithm to reduce delivery time. Their current average delivery time is 30 minutes. They claim that using the new algorithm will reduce delivery time to less than 30 minutes.

How to solve it (Step by step):

- Step 1: Define Hypotheses:

Null Hypothesis (H_0): Delivery time with the new algorithm ≥ 30 minutes (no improvement).

Alternative Hypothesis (H_1): Delivery time with the new algorithm < 30 minutes (improvement).

- Step 2: Determine Significance Level (α): Usually $\alpha = 0.05$ (5%), meaning we accept a 5% risk of error.
- Step 3: Data Collection: Data from 100 random deliveries using the new algorithm was collected. The average delivery time was 28 minutes with a standard deviation of 4 minutes.
- Step 4: Calculate Test Statistic: Since sample size is 100 (greater than 30), we use Z-test.

$$[Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{28 - 30}{4 / \sqrt{100}} = \frac{-2}{0.4} = -5]$$

- Step 5: Find P-value and make decision: For Z-score -5, the P-value is approximately 0.00000028, which is much smaller than 0.05.

Decision: Since $P\text{-value} < 0.05$, we reject the Null Hypothesis. This means the new algorithm has indeed reduced delivery time and should be implemented.

1. Interview Perspective

2. What is P-value? Explain in simple terms.

Sample Answer: P-value is the probability of obtaining the observed result, or something more extreme, assuming the Null Hypothesis is true. If the P-value is very small (usually less than 0.05), it means the Null Hypothesis is unlikely to be true.

The probability is very low, so we cancel it.

1. What are Type I and Type II Errors? Sample Answer: Type I Error (False Positive) occurs when we reject the Null Hypothesis even though it is true (e.g., punishing an innocent person). Type II Error (False Negative) occurs when we accept the Null Hypothesis even though it is false (e.g., letting a guilty person go).
2. What is the difference between Z-test and T-test? Sample Answer: When our sample size is large ($n > 30$) and the population standard deviation is known, we use the Z-test. When the

sample size is small ($n < 30$) and the population standard deviation is unknown, we use the T-test.

3. What is the difference between Null Hypothesis and Alternative Hypothesis? Sample

Answer: The Null Hypothesis is the default or current state (no change or no relationship). The Alternative Hypothesis is what we want to prove (there is a change or a relationship).

4. Why is the Significance Level (α) set at 0.05? Sample Answer: This is a standard practice. 0.05 means we accept a 5% risk that our decision might be wrong. However, in medical or sensitive cases, it is set at 0.01 (1%).

5. Beginner Mistakes

What mistakes do beginners make and how can they be avoided?

1. Considering the P-value directly as the “probability that the Alternative Hypothesis is true”: How to avoid: Remember that the P-value is the probability of obtaining the observed data assuming the Null Hypothesis is true.
2. Saying “Accept Null Hypothesis”: How to avoid: In statistics, we never “accept” the Null Hypothesis; we say “Fail to reject the Null Hypothesis” because we do not have enough evidence to reject it.
3. Confusing Type I and Type II Errors: How to avoid: Remember with real-life examples. Type I = False Alarm, Type II = Missed Detection.
4. Choosing the wrong test (using T-test instead of Z-test): How to avoid: Check the sample size and whether the population variance is known before selecting the test.
5. Confusing Correlation with Causation: How to avoid: Do not claim that one variable causes the other just because there is a correlation between them.

6. Formula

7. Z-Test Formula:

$$[Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}]$$

Explanation:

$(\bar{x})$ = Sample Mean

(μ) = Population Mean

(σ) = Population Standard Deviation

(n) = Sample Size

8. T-Test Formula:

$$[t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}]$$

Explanation:

(s) = Sample Standard Deviation (Others same as Z-test)

9. Practical Exercise

Beginner Exercise (Easy):

1. A company claims their bulbs last 1000 hours. You test 50 bulbs and find an average of 990 hours. The population standard deviation is 20 hours. Calculate the Z-score.
2. Is the average height of students in a class more than 65 inches? Height data of 10 students is given. Which test will you use and why?
3. If P-value is 0.03 and $\alpha = 0.05$, what will be your decision?

Intermediate Exercise (Moderate difficulty):

1. Daraz is testing two different colored "Add to Cart" buttons on their website (A/B Testing). Group A (red button) has a conversion rate of 5% and Group B (green button) has 6%. The sample size for each is 1000. Use a two-proportion Z-test to determine if this difference is significant.
2. To verify if a new training program improves employee skills, perform a paired T-test on the scores of 15 employees before and after training.

Real Business Exercise (Real company problem):

Problem: bKash has introduced a new feature in their app. They want to see if the average time users spend on the app has increased due to this feature. The previous average time was 5 minutes. After the new feature, data from 500 users shows an average time of 5.5 minutes with a standard deviation of 1.2 minutes.

Task: Set hypotheses, select the test, calculate, and provide a data-driven decision to management.

1. Career Perspective

Junior Data Analyst: Typically assists seniors by processing A/B Testing data using Excel or Python and calculating P-values.

Data Analyst: Determines which test to use, extracts data using SQL, runs tests with Python/R, and presents results to stakeholders.

Senior Data Analyst: Designs the entire experiment, determines sample size, conducts impact analysis of Type I/II errors, and supports the company's strategic decision-making.

Business Analyst: Focuses more on the business impact of results rather than statistical calculations. Seeks answers to questions like, "How much will the company's revenue increase due to this change?"

Summary and Cheat Sheet

Key Points:

- Hypothesis Testing is a mathematical method to verify a claim using data.
- Null Hypothesis (H0) is the default state; Alternative Hypothesis (H1) is what we want to prove.
- If P-value < 0.05, we reject H0.
- If P-value > 0.05, we fail to reject H0.

Quick Reference Cheat Sheet:

Condition	Which Test?
Sample Size > 30, Population σ known	Z-Test
Sample Size < 30, Population σ unknown	T-Test
Comparing means of two independent groups	Independent T-Test
Comparing means of the same group before and after	Paired T-Test
To examine relationships in categorical data	Chi-Square Test
Comparing means of more than two groups	ANOVA

This guide will help you become proficient in Hypothesis Testing from the basics to business applications.

Correlation (সহসম্পর্ক)

1. Definition

What is it? Correlation is a statistical measure that indicates how strongly two variables are related. It measures how one variable changes when the other changes. The value ranges from -1 to +1.

Simple explanation and everyday examples: In simple terms, correlation looks for relationships between two things. For example, the more you study, the better your exam results will be. Here, “study time” and “exam results” have a positive correlation. Conversely, the more fast food you eat, the worse your health becomes. Here, “fast food consumption” and “health” have a negative correlation.

2. Why is it important?

Why is it used in Data Analytics? Correlation is used in Data Analytics to find relationships between different variables in a dataset. It helps data scientists and analysts understand which factors influence each other. This allows for removing unnecessary variables and

making models more accurate (Feature Selection).

Importance in Business Decision-Making: Correlation is crucial in business decisions. It helps companies understand which marketing strategies work, which products are often sold together, or why customers leave. This enables companies to allocate resources effectively and increase profits.

3. How do companies use it?

E-commerce (e.g., Daraz, Amazon): When you buy a mobile phone on Amazon or Daraz, they suggest a phone cover or headphones. They use correlation to find that many who buy a phone also buy a cover. This is called cross-selling.

Banking (e.g., bKash, Brac Bank): Banks use correlation to analyze credit risk. For example, they observe a positive correlation between a customer's income and their ability to repay loans. Higher income generally means a lower chance of loan default.

Healthcare (e.g., Square Hospital): Hospitals analyze the correlation between patient age, weight, and various diseases (e.g., diabetes or heart problems). Through this, they understand which age or weight groups are at higher risk for certain diseases.

Telecom (e.g., Grameenphone): Grameenphone uses correlation to analyze customer churn rates. They observe that as the call drop rate increases, customers tend to switch to other operators more frequently (Positive Correlation).

Marketing (e.g., Digital Marketing Agencies): Marketing companies examine whether increasing Facebook ad spend leads to higher sales. By finding the correlation between Ad Spend and Sales, they determine their marketing budget.

Retail (e.g., Shwapno, Agora): Supermarkets analyze the correlation between weather and product sales. For example, umbrella and khichuri rice-lentil sales increase on rainy days. Based on this data, they manage inventory.

Manufacturing (e.g., Walton): Companies like Walton examine the correlation between machine maintenance and production quality. Regular maintenance reduces the rate of defective products (Negative Correlation).

1. How Does a Data Analyst Use It?

How to use it in Excel?

To find correlation in Excel, the CORREL function is used.

Formula: =CORREL(array1, array2)

Example: =CORREL(A2:A100, B2:B100) (where column A contains Ad Spend and column B contains Sales data).

How to use it in SQL?

Not all databases have a direct correlation function, but many modern databases (e.g.,

PostgreSQL, BigQuery) support the CORR() function.

Query:

```
SELECT CORR(ad_spend, sales_revenue) AS correlation_coefficient
FROM marketing_data;
```

How to use it in Python?

Using the Pandas library in Python, correlation can be easily calculated.

Code:

```
import pandas as pd

# Create DataFrame
data = {'Ad_Spend': [100, 200, 300, 400, 500],
        'Sales': [150, 250, 350, 450, 550]}
df = pd.DataFrame(data)

# Calculate Correlation
correlation = df['Ad_Spend'].corr(df['Sales'])
print(f"Correlation: {correlation}")
```

How to use it in Power BI?

In Power BI, correlation can be calculated using DAX.

DAX Measure:

```
Correlation =
VAR AvgX = AVERAGE(Table[Ad_Spend])
VAR AvgY = AVERAGE(Table[Sales])
VAR Numerator = SUMX(Table, (Table[Ad_Spend] - AvgX) * (Table[Sales] - AvgY))
VAR Denominator = SQRT(SUMX(Table, (Table[Ad_Spend] - AvgX)^2) * SUMX(Table,
RETURN DIVIDE(Numerator, Denominator)
```

1. Real Business Scenario

A real company problem: Suppose FoodPanda notices that as their delivery time increases, customer ratings decrease. They want to understand how strong this relationship is so they can decide whether to invest in reducing delivery time.

How to solve it (Step by step):

1. Data Collection: Collected delivery time (in minutes) and customer rating (1-5) data for the past month.
2. Data Cleaning: Removed incorrect or incomplete data.
3. Correlation Analysis: Calculated the correlation coefficient using Excel or Python. The correlation was found to be -0.85.
4. Interpretation: A value of -0.85 indicates a strong negative correlation, meaning as delivery time increases, ratings decrease.
5. Decision Making: Since the relationship is strong, FoodPanda decided to increase the number of delivery riders and improve route optimization algorithms to reduce delivery time and increase ratings.

6. Interview Perspective

Question 1: What is the difference between Correlation and Causation?

Sample Answer: Correlation means there is a relationship between two variables, but it does not imply that one causes the other (Causation). For example, ice cream sales and sunglasses sales may have a positive correlation, but ice cream sales do not cause sunglasses sales to increase. Both increase due to hot weather.

Question 2: What do the values -1, 0, and +1 of the Correlation Coefficient signify?

Sample Answer: +1

Meaning: Perfect positive correlation (when one increases, the other increases proportionally).

-1 means perfect negative correlation (when one increases, the other decreases proportionally).

0 means no linear relationship.

Question 3: How do outliers affect correlation?

Sample Answer:

Outliers can significantly alter the value of correlation. A large outlier can make a weak relationship appear strong or a strong relationship appear weak. Therefore, outliers should be checked before calculating correlation.

Question 4: What is the difference between Pearson and Spearman correlation?

Sample Answer:

Pearson correlation measures the linear relationship between two continuous variables. On the other hand, Spearman correlation is based on the ranks of the variables and can measure non-linear relationships as well.

Question 5: When should you not use correlation analysis?

Sample Answer:

When the relationship between two variables is non-linear (e.g., U-shaped), Pearson correlation does not provide accurate results. Also, if the data contains many outliers, correlation analysis should be avoided or outliers should be removed.

1. Beginner Mistakes

Mistake 1: Confusing correlation with causation.

How to avoid: Always remember that just because two things happen together does not mean one causes the other. Look for logical reasons.

Mistake 2: Calculating correlation without checking for outliers.

How to avoid: Before calculating correlation, create a scatter plot to visualize the data and check for outliers.

Mistake 3: Using Pearson correlation for non-linear relationships.

How to avoid: If the scatter plot shows a non-linear relationship (e.g., curved line), use Spearman correlation instead.

Mistake 4: Assuming correlation of 0 means no relationship.

How to avoid: A correlation of 0 means no linear relationship, but there may still be other complex (non-linear) relationships. Visualization is essential.

Mistake 5: Relying solely on the correlation coefficient.

How to avoid: Do not rely only on the number; always create graphs (scatter plots) to understand the data pattern.

1. Formula

Pearson Correlation Coefficient ®:

$$[r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}]$$

Explanation of each part/symbol in the formula:

- r : Correlation coefficient.
 - (x_i) : Each data point of the first variable (x).
 - (y_i) : Each data point of the second variable (y).
 - $(\bar{x})$: Mean of the first variable (x).
 - $(\bar{y})$: Mean of the second variable (y).
 - $(\sum)$: Summation or total sum.
-

1. Practical Exercise

3 Beginner Exercises:

1. Exercise: Calculate the correlation in Excel using the data below.

Reading Time (hours): 2, 3, 5, 7, 9

Exam Marks: 50, 60, 75, 85, 95

2. Exercise: Using Python, calculate the correlation between Age and Height.

Age: 10, 12, 14, 16, 18

Height (cm): 130, 140, 155, 165, 170

3. Exercise: Draw a scatter plot and determine whether the relationship is positive or negative.

Temperature: 20, 25, 30, 35, 40

Winter Clothing Sales: 100, 80, 50, 20, 5

2 Intermediate Exercises:

1. Exercise: Add an outlier to a dataset and observe how the correlation coefficient changes.
2. Exercise: Using SQL, calculate the correlation between product price and quantity sold in an e-commerce database.

1 Real Business Exercise:

Problem: In the “Pathao” ride-sharing app, it is observed that the ride cancellation rate increases on rainy days. Obtain data on rainfall (mm) and cancellation rate (%) for the last 30 days. Use Excel or Python to analyze.

Calculate the correlation and provide management with a report on how strong the relationship is between rain and cancellations, and what can be done to address this (for example: giving bonuses to riders on rainy days).

1. Career Perspective

Junior Data Analyst: A junior analyst typically uses Excel or Python to find correlations between variables in a dataset and creates scatter plots to report to seniors.

Data Analyst: A data analyst uses correlation for feature selection, i.e., determining which variables are important for predictive modeling, and displays insights on dashboards (Power BI/Tableau).

Senior Data Analyst: Senior analysts analyze correlation results to develop business strategies. They understand the difference between causation and correlation and solve complex business problems.

Business Analyst: A business analyst uses correlation to understand which business metrics are related to each other. For example, by analyzing the relationship between marketing campaigns and sales growth, they advise management on budget optimization.

Summary and Cheat Sheet

Main Points:

- Correlation indicates the degree and direction of the relationship between two variables.
- The value ranges from -1 (perfect negative) to +1 (perfect positive). A value of 0 means no linear relationship.
- Correlation does not imply causation.
- Outliers can significantly affect correlation.

Quick Reference Cheat Sheet:

- Positive Correlation (+): When one increases, the other also increases (e.g., ad spend and sales).
- Negative Correlation (-): When one increases, the other decreases (e.g., product price and demand).
- No Correlation (0): No relationship (e.g., shoe size and IQ).
- Excel Formula: `=CORREL(array1, array2)`
- Python Code: `df['A'].corr(df['B'])`
- SQL Query: `SELECT CORR(col1, col2) FROM table;`
- Golden Rule: "Correlation does not imply Causation."

Regression

1. Definition

What is it? Regression is a statistical method used to determine the relationship between a dependent variable and one or more independent variables. Simply put, it is a mathematical way to find out how the change in one factor affects another.

Explanation in simple terms and everyday example: Suppose you want to know how your study time (in hours) affects your exam scores (marks). Here, 'exam score' is the dependent variable because it depends on how long you study. 'Study time' is the independent variable. Using regression, you can understand how much your score might increase on average if you study one extra hour daily.

Another example: Suppose you run an ice cream shop. You notice that ice cream sales increase as temperature rises. By performing regression analysis, you can find out exactly how many more ice creams are sold for every 1 degree Celsius increase in temperature.

2. Why is it important?

Why is it used in Data Analytics? Regression is very important in data analytics because it

not only tells whether there is a relationship between two factors but also determines the magnitude and direction of that relationship. It helps forecast the future by analyzing past data.

What is its importance in Business Decision-making? Regression is essential for making accurate business decisions. Companies use it to understand which factors most affect their sales or profits. For example, it can predict how much sales will increase if a certain amount is spent on marketing. This enables proper budgeting, resource allocation, and risk management.

3. How do companies use it?

Real-world examples:

- E-commerce (e.g., Daraz, Amazon): Regression models are built using customer browsing history, age, and past purchase data to predict which product a customer is likely to buy in the future.

You can buy or how much money you can spend.

Banking (e.g., bKash, BRAC Bank): Regression is used for credit scoring. Based on the customer's income, loan history, and profession, the probability of loan default is determined.

Healthcare (e.g., Square Hospital): By analyzing the patient's age, weight, blood pressure, and lifestyle data, the risk of developing a specific disease in the future (e.g., diabetes or heart disease) is predicted.

Telecom (e.g., Grameenphone): By analyzing the customer's call duration, internet usage, and recharge patterns, it is estimated which customer might switch to another operator (Customer Churn).

Marketing (e.g., Unilever): Based on advertising expenses on TV, social media, and newspapers, regression is used to determine how much product sales will increase (Marketing Mix Modeling).

Retail (e.g., Shwapno): Based on holidays, weather, and promotional offers, demand forecasting is done to determine the demand for certain products in the upcoming week.

Manufacturing (e.g., Walton): By analyzing machine age, usage time, and maintenance history, predictive maintenance estimates when a machine might break down.

1. How Does a Data Analyst Use It?

How to use it in Excel? Regression can be performed in Excel using the LINEST function or the Data Analysis Toolpak.

Formula: =LINEST(known_y's, known_x's, [const], [stats])

Example: If sales (Y) are in column A and marketing expenses (X) are in column B, then =LINEST(A2:A100, B2:B100) can be used to find the regression coefficient.

How to use it in SQL? SQL is not ideal for directly building regression models but is used to prepare data and calculate the slope of a simple linear regression. Example query:

```
SELECT
    (COUNT(*)*SUM(x*y) - SUM(x)*SUM(y)) / (COUNT(*)*SUM(x*x) - SUM(x)*SUM(x))
FROM sales_data;
```

How to use it in Python? Regression can be easily done in Python using libraries like scikit-learn or statsmodels. Example code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Load data
data = pd.read_csv('sales_data.csv')
X = data[['Marketing_Spend']] # Independent Variable
y = data['Sales'] # Dependent Variable

# Create and fit model
model = LinearRegression()
model.fit(X, y)

print("Coefficient (Slope):", model.coef_)
print("Intercept:", model.intercept_)
```

How to use it in Power BI? In Power BI, regression can be visualized by adding a Trend Line to a Scatter Chart. Additionally, linear regression can be calculated using DAX. Example DAX formula to calculate the slope:

```
Slope =
VAR AvgX = AVERAGE(Sales[Marketing_Spend])
VAR AvgY = AVERAGE(Sales[Revenue])
RETURN
DIVIDE(
    SUMX(Sales, (Sales[Marketing_Spend] - AvgX) * (Sales[Revenue] - AvgY))
    SUMX(Sales, (Sales[Marketing_Spend] - AvgX)^2)
)
```

1. Real Business Scenario

A real company problem: Suppose 'FoodPanda' wants to understand how delivery fee and discount affect their daily order numbers. They are uncertain about what delivery fee to set for the upcoming month.

Step-by-step solution process:

2. Data Collection: Collected daily data for the last 6 months on delivery fee, discount amount, and total orders.
3. Data Cleaning: Removed missing values and outliers (abnormal data).
4. Model Selection: Since there are two independent variables (Delivery Fee and Discount) and one dependent variable (Total Orders), Multiple Linear Regression will be used.
5. Model Training: Ran the regression model on the data using Python or Excel.
6. Interpretation: The model showed:
 - The coefficient for Delivery Fee is -500, meaning if the delivery fee increases by 1 Taka, daily orders decrease by 500.
 - The coefficient for Discount is +800, meaning if the discount increases by 1 Taka, daily orders increase by 800.
7. Decision Making: Using these insights, FoodPanda decided to slightly reduce the delivery fee and increase discounts at specific restaurants to maximize their total orders and profit.
You can.

8. Interview Perspective

Interview Questions and Sample Answers:

1. Question: What is the difference between Correlation and Regression?

Answer: Correlation only measures the type and strength of the relationship between two variables. However, Regression helps predict or estimate the value of one variable based on another variable and attempts to explain cause-and-effect relationships.

2. Question: What is R-squared (R^2) and why is it important?

Answer: R-squared is a metric that indicates how well our model explains the data. Its value ranges between 0 and 1. An $R^2 = 0.80$ means that 80% of the variation in the dependent variable can be explained by the independent variable(s).

3. Question: What are the main assumptions of Linear Regression?

Answer: The main assumptions are: Linearity (the relationship must be linear), Independence (observations must be independent), Homoscedasticity (variance must be constant), and Normality (errors or residuals must follow a normal distribution).

4. Question: What is Multicollinearity and why is it harmful?

Answer: When multiple independent variables in a regression model are highly correlated with each other, it is called Multicollinearity. It is harmful because it makes it difficult for the model to determine the actual impact of each variable on the dependent variable, reducing the model's reliability.

5. Question: What is Overfitting and how can it be avoided in regression?

Answer: Overfitting occurs when a model performs very well on training data but poorly on new data. To avoid it, we can use Regularization techniques (such as Ridge or Lasso Regression) or remove unnecessary variables.

9. Beginner Mistakes

Common mistakes beginners make and how to avoid them:

1. Mistake: Assuming Correlation implies Causation.

How to avoid: Regression does not mean that X causes Y. Decisions must be made considering logical and business context.

2. Mistake: Running the model without checking for Outliers.

How to avoid: Before running the model, identify outliers using Scatter Plots or Box Plots and handle them (remove or transform).

3. Mistake: Blindly relying on R-squared value.

How to avoid: Do not rely solely on R-squared; also check Adjusted R-squared, P-values, and Residual Plots.

4. Mistake: Using Categorical Data (e.g., gender, city names) directly in the model.

How to avoid: Convert categorical data into One-Hot Encoding or Dummy Variables before inputting into the model.

5. Mistake: Not checking assumptions.

How to avoid: After running the model, perform Residual Analysis to verify if assumptions like Linearity and Homoscedasticity are met.

10. Formula

Simple Linear Regression Formula:

$$[Y = \beta_0 + \beta_1 X + \epsilon]$$

Explanation of components:

- Y (Dependent Variable): The dependent variable whose value we want to predict (e.g., Sales).

- X (Independent Variable): The independent variable used for prediction (e.g., Marketing Spend).
- β_0 (Y-Intercept): The intercept; the value of Y when $X = 0$.
- β_1 (Slope/Coefficient): The slope; how much Y changes when X changes by one unit.
- ϵ (Error Term): The error or residual; the amount the model could not predict.

1. Practical Exercise

Beginner Exercise (Easy):

1. Exercise 1: In Excel, given study hours (2, 3, 4, 5, 6 hours) and scores (50, 60, 70, 80, 90) for 5 students, create a Scatter Plot and add a Trendline.
2. Exercise 2: Using the above data, find β_1 and β_0 in Excel using the `=SLOPE()` and `=INTERCEPT()` functions.
3. Exercise 3: If a student studies for 7 hours, calculate their probable score using the regression equation.

Intermediate Exercise (Moderate difficulty):

1. Exercise 4: Using Python's scikit-learn, load a dummy dataset (e.g., House Price Prediction), create a Simple Linear Regression model, and find the R-squared value.
2. Exercise 5: Add an outlier to a dataset and observe how the presence of the outlier affects the Regression Line.

How it changes. Then remove outliers and run the model again.

Real Business Exercise (Problem of a Real Company):

1. Exercise 6: You are a data analyst at 'Pathao'. You have been given data for the past 1 year on 'Rainfall Amount (mm)', 'Traffic Jam Index', and 'Ride Cancellation Rate (%)'. Using Python or Excel, create a Multiple Regression model and determine which factor is most responsible for ride cancellations.

2. Career Perspective

Junior Data Analyst: Typically runs basic regression models on cleaned data using Excel or Python, checks R-squared or P-value, and prepares reports.

Data Analyst: Performs data cleaning, feature engineering, and checks model assumptions. They present regression results in simple language to business stakeholders (e.g., "Increasing marketing budget by 10% will increase sales by 5%").

Senior Data Analyst: Builds complex Multiple Regression or Logistic Regression models. They optimize model performance and directly contribute to the company's strategic decision-making.

Business Analyst: Focuses more on the business impact of regression results than coding. Using model predictions, they develop new business strategies, pricing models, or marketing plans.

Summary and Cheat Sheet

Summary: Regression is a statistical tool that helps us understand how one variable (e.g., sales) depends on one or more other variables (e.g., advertising expenses, price). It is widely used to analyze past data for forecasting and making data-driven business decisions.

Quick Reference Cheat Sheet:

- **Dependent Variable (Y):** The variable to predict (Target).
- **Independent Variable (X):** The variable(s) used to predict (Predictor).
- **Simple Linear Regression:** One X and one Y ($Y = \beta_0 + \beta_1 X$).
- **Multiple Linear Regression:** Multiple Xs and one Y ($Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots$).
- **Logistic Regression:** When Y is categorical (e.g., Yes/No, Spam/Not Spam).
- **R-squared (R^2):** Model accuracy or goodness of fit. The closer to 1, the better the model.
- **P-value:** Tests whether the independent variable truly affects the dependent variable. If $P\text{-value} < 0.05$, the variable is significant (Statistically Significant).
- **Coefficient (β):** The amount Y changes when X changes by one unit. Positive means positive relationship; negative means negative relationship.

Sampling (Sample Selection)

1. Definition

What is it? Sampling is a statistical method where a small part or sample is selected from a large dataset or population so that analyzing this small part can provide insights or conclusions about the entire population.

Explanation in simple terms with a daily life example: Suppose you are cooking a pot of biryani. To check if the biryani is properly cooked or salted correctly, you don't eat the whole pot. Instead, you take just one spoonful to test. This one spoonful is your "Sample," and the whole pot is the "Population." By tasting one spoon, you can decide about the whole pot's flavor. The same principle applies in data analytics.

2. Why is it important?

Why is it used in Data Analytics?

- **Saves time and cost:** Working with millions or billions of data points is time-consuming and expensive. Sampling allows working with less data to get faster results.

- Easier analysis: Processing large datasets requires significant computational power. Using sample data makes analysis possible on ordinary computers.
- Accuracy: If the sample is selected correctly, it can accurately represent the entire population.

Importance in Business Decision-making: Business leaders need to make quick decisions. For example, before launching a new product, it is impossible to gather opinions from the entire country. Instead, they conduct surveys on a specific sample group and make product launch decisions based on those results.

This reduces risk and allows for quicker market entry.

1. How do companies use it? E-commerce (e.g., Daraz, Amazon): To understand customer satisfaction, they do not call every customer. Instead, they randomly select a sample of customers and gather their feedback to estimate the satisfaction rate of the entire customer base. Banking (e.g., bKash, Brac Bank): It is impossible to manually check millions of daily credit card transactions for fraud. They take a sample of suspicious transactions, analyze them deeply, and identify fraud patterns. Healthcare (e.g., Square Hospital): For testing the efficacy of a new drug (Clinical Trial), the drug is not given to all patients in the country. It is applied to a specific sample of patients, and the results are observed. Telecom (e.g., Grameenphone): To check network quality, they analyze the call drop rate in selected areas (sample) to get an idea of the network performance across the country. Marketing (e.g., Unilever): To understand the effectiveness of a new advertisement, they run the ad on a small focus group (sample) and observe their reactions before launching the main campaign. Retail (e.g., Shwapno): During inventory or stock audits, they do not count all products in the warehouse but count products on selected shelves (sample) to estimate the total stock. Manufacturing (e.g., Walton): For quality control in factories producing thousands of fridges or TVs, a random sample of products is checked.

2. How does a Data Analyst use it? How to use it with Excel? Sampling can be done using the Data Analysis Toolpak or formulas in Excel.

Function/Formula: =INDEX(A2:A100, RANDBETWEEN(1, COUNTA(A2:A100)))

This formula randomly selects one data point from cells A2 to A100.

How to use it with SQL? To take a random sample from a large table in SQL, ORDER BY RAND() or TABLESAMPLE is used.

Query example (MySQL):

```
SELECT * FROM customers
ORDER BY RAND()
LIMIT 100;
```

This will fetch data of 100 random customers from the customers table.

How to use it with Python? Sampling can be easily done using Python's Pandas library.

Code example:

```
import pandas as pd

# Load dataset
df = pd.read_csv('sales_data.csv')

# Randomly take 10% of data as sample
sample_df = df.sample(frac=0.1, random_state=42)
print(sample_df.head())
```

How to use it with Power BI? Although Power BI has limited built-in sampling functions, sampling can be done using DAX or Power Query.

DAX/Measure example:

RandomSample = CALCULATE(

```
SUM(Sales[Amount]),
FILTER(Sales, RAND() ≤ 0.1) -- 10% random sample
```

)

1. Real Business Scenario A real company problem: Suppose “FoodPanda” is receiving customer complaints about their delivery time. They want to know the average delivery time in Dhaka city. However, millions of deliveries happen daily, and processing all data is time-consuming.

How sampling is used to solve the problem (Step by step):

1. Define Population: All delivery data in Dhaka city over the past month (assume 1 million).
2. Determine Sample Size: Using statistical formulas, they decide that data from 5,000 deliveries will provide accurate results.
3. Select Sampling Method: They use Simple Random Sampling to randomly select 5,000 records from the 1 million data points.
4. Analysis: The average delivery time for these 5,000 deliveries is calculated and found to be 45 minutes.
5. Decision: Based on this sample, management decides to hire more riders or optimize routes to reduce delivery time.

6. Interview Perspective

7. What is the difference between Population and Sample?

Sample Answer: Population is the entire dataset or all elements, while a Sample is a smaller part selected from that Population. For example, the entire country's population is the Population, and 1,000 people selected for a survey is the Sample.

8. What is the difference between Simple Random Sampling and Stratified Sampling?

Sample Answer:

Simple Random Sampling gives each element of the population an equal chance of being selected (e.g., lottery). On the other hand, Stratified Sampling divides the population into several groups (Strata) and randomly samples from each group (e.g., sampling separately from male and female groups).

1. What is Sampling Bias and how can it be avoided?

Sample Answer: Sampling Bias is a situation where the sample does not accurately represent the entire population. To avoid this, ensure randomization and select an appropriate sample size.

2. How to determine Sample Size?

Sample Answer: Sample size determination depends on Population Size, Confidence Level

(usually 95%), and Margin of Error (usually 5%). Specific statistical formulas or online calculators are used for this purpose.

3. Why is the Central Limit Theorem (CLT) important in sampling?

Sample Answer: CLT states that as the sample size increases (usually above 30), the distribution of the sample mean approaches a normal distribution, regardless of the population's distribution. This helps in hypothesis testing.

1. Beginner Mistakes

What mistakes do beginners commonly make?

1. Taking a very small sample size: This leads to inaccurate results and increased margin of error.
2. Non-random sampling: Selecting samples based on convenience, which introduces bias.
3. Improperly defining the population: Not clearly specifying who will be surveyed.
4. Ignoring outliers: Overlooking abnormal data points within the sample.
5. Treating sample results as 100% certain: A sample always provides an estimate, never a perfect representation of the population.

How to avoid these?

- Use statistical formulas to determine the correct sample size.
- Always try to use random sampling methods.
- Check for outliers during data cleaning.
- Report confidence intervals and margin of error along with results.

1. Formula

The most important formula in sampling is the Sample Size determination formula (Cochran's Formula):

$$n = \frac{Z^2 \cdot p \cdot (1 - p)}{e^2}$$

Explanation of each part/symbol of the formula:

n = Sample Size (number of data points needed).

Z = Z-score (based on confidence level, e.g., for 95% confidence level Z -score is 1.96).

p = Estimated proportion (estimated percentage of the population with a particular attribute).

e = Margin of Error (acceptable error level, e.g., 5% or 0.05).

1. Practical Exercise

Beginner Exercise (Easy):

1. Excel Exercise: You have marks of 100 students. Use Excel's RANDBETWEEN function to create a random sample of 10 students.
2. Python Exercise: From the list [10, 20, 30, 40, 50, 60, 70, 80, 90, 100], use Python's random.sample() to select 3 random numbers.
3. SQL Exercise: From an employees table, use ORDER BY RAND() LIMIT 5 to retrieve data of 5 employees.

Intermediate Exercise (Moderate difficulty):

1. Stratified Sampling (Python): Given a dataset with 'Male' and 'Female' genders, use Pandas to sample data proportionally from both genders.
2. Sample Size Calculation: Calculate sample size for an unknown population with 99% confidence level and 3% margin of error (Z -score = 2.576).

Real Business Exercise (Real company problem):

Problem: "Daraz" wants to test their new website design (A/B Testing). They have 100,000 visitors daily.

Task: As a data analyst, decide how many visitors should see the new design and how many should see the old design.

1. Career Perspective

Junior Data Analyst: Typically works under seniors using SQL or Python to handle large databases.

Works on extracting sample data from the source.

Data Analyst: Determines which type of Sampling Method (Random, Stratified, Cluster) to use and performs EDA (Exploratory Data Analysis) on the sample data to derive insights.

Senior Data Analyst: Identifies sampling bias, designs A/B testing, and builds predictive models based on the sample data.

Business Analyst: Transforms insights obtained from sample data into business strategies. For example, advises where to allocate the marketing budget based on the results of a sample survey.

Summary and Cheat Sheet

Key points summarized:

- Sampling: Selecting a small portion from a large population.
- Purpose: To save time, cost, and computational power.
- Types: Probability Sampling (Random, Stratified) and Non-probability Sampling (Convenience, Quota).
- Importance: Proper sampling captures an accurate picture of the entire population, aiding business decision-making.

Quick Reference Cheat Sheet:

Concept	Description (বিবরণ)	Example (উদাহরণ)
Population	The entire dataset or whole group.	All internet users in Bangladesh.
Sample	A small portion taken from the population.	1,000 users selected for a survey.
Simple Random	Everyone has an equal chance of selection, like a lottery.	SQL: ORDER BY RAND()
Stratified	Dividing into groups and sampling from each group.	Sampling separately from male and female groups.
Sample Size (n)	The number of data points to be taken.	Calculated using Cochran's Formula.
Bias	When the sample does not accurately represent the population.	Making nationwide decisions based only on opinions from Dhaka city residents.

Standard Deviation (প্রমাণ বিচ্যুতি)

1. Definition

What is it? Standard Deviation is a statistical measure that indicates how much the values in a dataset deviate or spread out from their mean. Simply put, it measures the “spread” or “dispersion” of data.

Explanation in simple Bengali and everyday example: Suppose you are tracking how long it takes you to get to the office each day.

- First week: Monday 30 minutes, Tuesday 32 minutes, Wednesday 28 minutes, Thursday 31 minutes, Friday 29 minutes. Average time = 30 minutes. Here, daily times are very close to the average, so Standard Deviation is low. You can be confident that it will take about 30 minutes to get to the office.
- Second week: Monday 20 minutes, Tuesday 45 minutes, Wednesday 15 minutes, Thursday 50 minutes, Friday 20 minutes. Average time = 30 minutes. But here, there is a large variation in daily times. Some days you arrive very early, other days very late. Hence, Standard Deviation is high. You cannot be sure how long it will take to get to the office on any given day.

A low Standard Deviation means data points are close to the mean (consistent). A high Standard Deviation means data points are spread far from the mean (inconsistent/volatile).

1. Why is it important?

Why is it used in Data Analytics?

- Verifying data reliability: The mean does not always represent the true picture. Standard Deviation shows how reliable the data is.
- Outlier detection: Helps identify if there are any abnormal values (outliers) in the dataset.
- Risk measurement: Used in finance and analytics to measure data volatility or risk.

Importance in Business Decision-making:

- Quality Control: To determine if the quality of produced goods is consistent.
- Inventory Management: To understand fluctuations in product demand and manage stock accordingly.
- Performance Evaluation: To assess how consistent the performance of employees or campaigns is.

1. How do companies use it? Real-life examples:

E-commerce (e.g., Daraz): Calculates the Standard Deviation of delivery times. If it is high, it means customers are receiving deliveries at varying times, which can cause customer dissatisfaction. Banking (e.g., bKash, BRAC Bank): Monitors the Standard Deviation of customers' transaction amounts to detect unusual transactions or fraud. Healthcare (e.g., Square Hospital): Analyzes patients' blood pressure or heart rate data. A high Standard Deviation indicates the patient's condition is unstable. Telecom (e.g., Grameenphone): Measures the Standard Deviation of call drop rates or internet speeds. A high deviation in speed is considered a sign of poor network quality. Marketing (e.g., Google Ads): Reviews the Standard Deviation of conversion rates across various ad campaigns. Campaigns with low deviation and good average performance receive higher budgets.

Retail (e.g., Shwapno): Examines the Standard Deviation of daily sales. High sales fluctuations prompt extra caution in inventory management. Manufacturing (e.g., Walton): Checks the Standard Deviation of measurements of produced fridge or AC parts. High deviation indicates machine faults (Quality Control).

2. How does a Data Analyst use it? How to use it in Excel?

Population Standard Deviation: =STDEV.P(A2:A100) (when the entire dataset is available)

Sample Standard Deviation: =STDEV.S(A2:A100) (when only a part or sample of the dataset is available)

How to use it in SQL?

```
-- Sample Standard Deviation
SELECT STDDEV_SAMP(sales_amount) AS std_dev_sales
FROM daily_sales;

-- Population Standard Deviation
SELECT STDDEV_POP(sales_amount) AS pop_std_dev_sales
FROM daily_sales;
```

How to use it in Python?

```

import pandas as pd
import numpy as np

# Create DataFrame
data = {'Sales': [100, 120, 110, 150, 130, 90, 140]}
df = pd.DataFrame(data)

# Sample Standard Deviation (Pandas default is sample, ddof=1)
std_dev = df['Sales'].std()
print(f"Standard Deviation: {std_dev}")

# Population Standard Deviation (Numpy default is population, ddof=0)
pop_std_dev = np.std(df['Sales'])
print(f"Population Standard Deviation: {pop_std_dev}")

```

How to use it in Power BI? Using DAX (Data Analysis Expressions):

- Sample Standard Deviation Sales_StdDev = STDEV.S(Sales[SalesAmount])
- Population Standard Deviation Sales_StdDev_Pop = STDEV.P(Sales[SalesAmount])

1. Real Business Scenario A real company problem: Suppose Foodpanda has two riders (Rider A and Rider B). Their delivery times (in minutes) over the last 5 days are as follows:

Rider A: 30, 32, 28, 31, 29 Rider B: 15, 45, 20, 50, 20

The manager wants to know which rider is more reliable and to whom important deliveries should be assigned.

Solution (Step by step): Step 1: Calculate the mean

Rider A's mean = $(30+32+28+31+29) / 5 = 30$ minutes

Rider B's mean = $(15+45+20+50+20) / 5 = 30$ minutes

From the mean, both seem equally skilled.

Step 2: Calculate Standard Deviation

Rider A's Standard Deviation = ~1.58 minutes

Rider B's Standard Deviation = ~16.58 minutes

Step 3: Decision Making Rider A's Standard Deviation is much lower (1.58), meaning he consistently delivers close to 30 minutes. He is very consistent and reliable. On the other hand, Rider B's Standard Deviation is much higher (16.58), meaning his delivery times vary widely; sometimes very fast, sometimes very late. He is unpredictable. Decision: The manager should consider Rider A more reliable and assign important deliveries to him.

1. Interview Perspective Question 1: What is the difference between Standard Deviation and Variance? Sample Answer: Variance is the average of the squared differences from the mean. Standard Deviation is the square root of the Variance. Standard Deviation is in the same unit as the original data (e.g., currency, kg), making it easier to interpret. Variance, however, is in squared units.

Question 2: What is the difference between Population and Sample Standard Deviation?

Sample Answer: When...

When we have the entire dataset (Population), we use Population Standard Deviation (where division is by N). But when we work with a small part of the entire dataset (Sample), we use Sample Standard Deviation (where division is by N-1, known as Bessel's correction, to reduce bias).

Question 3: What does it mean if the Standard Deviation is zero?

Sample Answer: A Standard Deviation of zero means every value in the dataset is equal to the mean. In other words, there is no variation or change in the data. All values are exactly the same.

Question 4: How do outliers affect the Standard Deviation?

Sample Answer: Standard Deviation is very sensitive to outliers. If there is a large outlier in the dataset, it changes the mean and as a result, the Standard Deviation increases significantly, which may misrepresent the data.

Question 5: When would you use IQR (Interquartile Range) instead of Standard Deviation?

Sample Answer: When the dataset contains many outliers or is skewed, Standard Deviation does not provide an accurate picture. In such cases, using IQR to measure data spread is more reasonable because IQR is not affected by outliers.

1. Beginner Mistakes

What mistakes do beginners commonly make and how to avoid them?

2. Mistake: Confusing Sample and Population Standard Deviation.

How to avoid: If the data is the entire population, use STDEV.P (or divide by N). If it is a sample, use STDEV.S (or divide by N-1).

3. Mistake: Making decisions based solely on the mean.

How to avoid: Always check the Standard Deviation along with the mean. Even if the means are the same, the data spread can differ.

4. Mistake: Calculating Standard Deviation without checking for outliers.

How to avoid: Visualize the data (e.g., using a boxplot) to check for outliers first. If necessary, remove outliers or use IQR for analysis.

5. Mistake: Directly comparing Standard Deviations with different units.

How to avoid: When comparing variation between two datasets with different units (e.g., one in currency, another in kilograms), use the Coefficient of Variation (CV = Standard Deviation / Mean).

6. Mistake: Confusing Standard Deviation with Variance.

How to avoid: Remember, Variance is the squared value. Taking the square root of Variance gives the Standard Deviation, which is in the original unit of the data.

1. Formula

Population Standard Deviation (σ - Sigma):

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Sample Standard Deviation (s):

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Explanation of each part/symbol in the formula:

- σ (Sigma): Population Standard Deviation.
 - s: Sample Standard Deviation.
 - (x_i) : Each individual value in the dataset.
 - μ (Mu): Population Mean (mean of the entire dataset).
 - $(\bar{x})$ (x-bar): Sample Mean (mean of the sample).
 - $(\sum)$ (Sigma): Summation (sum of).
 - N: Total number of data points in the population.
 - n: Total number of data points in the sample.
 - $n - 1$: Degrees of freedom (Bessel's correction).
-

1. Practical Exercise

Beginner Exercise (Easy):

1. Exercise 1: Scores of 5 students in Mathematics: 70, 75, 80, 85, 90. Calculate the Sample Standard Deviation.
2. Exercise 2: Sales of a shop over 4 days (in currency): 1000, 1000, 1000, 1000. What is the Standard Deviation? (Answer: 0, because there is no variation).
3. Exercise 3: Using Excel, calculate the Population Standard Deviation for the data: 12, 15, 18, 20, 22.

Intermediate Exercise (Moderate difficulty):

1. Exercise 4: The last 5 days' share prices of two companies are given.
Company X: 50, 52, 49, 51, 50
Company Y: 30, 70, 20, 80, 50
Which company's shares have higher risk? Prove using Python or Excel.
2. Exercise 5: Write an SQL query to calculate the Standard Deviation of the salary column from an employees table and explain how outliers would affect it.

Do it.

Real Business Exercise (Real Company Problem):

Exercise 6: As an analyst at Pathao, you have been given the data on 'Waiting Time' for rides in Dhaka city over the past month. You observe that the average waiting time is 10 minutes, but the Standard Deviation is 8 minutes. Based on this data, what insights would you provide to management, and what recommendations would you make to improve customer experience?

1. Career Perspective

- Junior Data Analyst: Primarily uses Excel or SQL to calculate the Standard Deviation of datasets and presents it in reports or dashboards (Power BI/Tableau).
- Data Analyst: Uses Standard Deviation to identify outliers in data, performs data cleaning, and analyzes data variance during A/B Testing.
- Senior Data Analyst: Uses it to understand data distribution when building various predictive models. Explains data volatility in major company decisions (e.g., new product launches or risk management).
- Business Analyst: Investigates the business logic behind the data. If Standard Deviation is high, identifies why there is so much inconsistency in the process and develops strategies for process improvement.

Summary and Cheat Sheet

Key points summarized:

- Standard Deviation measures how far data spreads from its mean.
- Low Standard Deviation = data is consistent and close to the mean.
- High Standard Deviation = data is inconsistent and far from the mean.
- It is widely used to detect outliers and measure risk or volatility.
- For sample data, division is by N-1; for population data, division is by N.

Quick Reference Cheat Sheet:

Concept	Formula / Function	Use Case
Population Std Dev	Excel: =STDEV.P()	When the entire dataset is available.
	SQL: STDDEV_POP()	
	Python: np.std()	
Sample Std Dev	Excel: =STDEV.S()	When only a sample of the dataset is used.
	SQL: STDDEV_SAMP()	
	Python: df.std()	
Variance	(Standard Deviation) ²	Measures data spread (in squared units).
Low Std Dev	Data is clustered around mean	Consistent performance, low risk.
High Std Dev	Data is spread out	Inconsistent performance, high risk.
Outlier Impact	Highly sensitive	Presence of outliers greatly increases Std Dev.

Variance (বিস্তার)

1. Definition

What is it? Variance is a statistical measure that specifies how far the values in a dataset

spread from their mean. Simply put, it measures how variable or dispersed the data is.

Explanation in simple terms with a daily life example: Suppose you are tracking how long it takes you to get to the office each day.

- Week 1: Monday 30 minutes, Tuesday 30 minutes, Wednesday 30 minutes, Thursday 30 minutes, Friday 30 minutes. Average time = 30 minutes. Here, there is no variation in time, so Variance = 0.
- Week 2: Monday 20 minutes, Tuesday 40 minutes, Wednesday 15 minutes, Thursday 45 minutes, Friday 30 minutes. Average time = 30 minutes. But the daily times vary significantly. This difference or dispersion from the mean is measured by Variance.
- High Variance means there is more uncertainty or risk in the data.

1. Why is it important?

Why is it used in Data Analytics? Decisions cannot be made accurately by looking at the mean alone because two different datasets can have the same mean but very different internal distributions. Variance helps us understand whether the data is consistent or not. It is crucial for identifying outliers and evaluating the performance of machine learning models.

What is its importance in Business Decisions? In business, Variance signifies risk or uncertainty. Uncertainty.

Risk Management: Stocks in the share market with high variance experience high risk.
Quality Control: Low variance in production means the product quality is consistent.
Sales Forecasting: High variance in sales makes inventory management difficult.

1. How do companies use it? E-commerce (e.g., Daraz, Amazon): Variance in delivery times is measured. If the variance is very high, it means customers sometimes receive products very quickly and other times very late, causing customer dissatisfaction. Banking (e.g., bKash, City Bank): Variance in customers' transaction amounts is analyzed for fraud detection. Sudden abnormal variance in an account is flagged as suspicious. Healthcare (e.g., Square Hospital): Variance in patients' blood pressure or sugar levels is used to measure medication effectiveness. Telecom (e.g., Grameenphone): Variance in call drop rates or internet speed is measured. High variance in speed indicates the network is not stable. Marketing (e.g., Google Ads): Variance in conversion rates of different ad campaigns helps identify which campaigns are consistently performing well. Retail (e.g., Shwapno): Daily sales variance is analyzed for stock or inventory management. Manufacturing (e.g., Walton): Variance in the

weight or size of produced goods is measured. High variance indicates possible machine issues and the need to strengthen quality control.

2. How does a Data Analyst use it? How to use it in Excel? There are two main functions to calculate variance in Excel:

Population Variance (when the entire dataset is available): =VAR.P(A2:A100)

Sample Variance (when only a part of the data is available): =VAR.S(A2:A100)

How to use it in SQL? SQL also has functions to calculate population and sample variance:

```
-- Sample Variance
SELECT VAR_SAMP(sales_amount) AS sample_variance FROM sales_data;

-- Population Variance
SELECT VAR_POP(sales_amount) AS population_variance FROM sales_data;
```

How to use it in Python? Using Python's pandas or numpy libraries, variance can be easily calculated:

```
import pandas as pd
import numpy as np

data = [10, 20, 30, 40, 50]
df = pd.DataFrame(data, columns=['Sales'])

# Using Pandas (Sample Variance by default)
variance_pandas = df['Sales'].var()

# Using Numpy (Population Variance by default)
variance_numpy = np.var(data)

print(f"Sample Variance: {variance_pandas}")
print(f"Population Variance: {variance_numpy}")
```

How to use it in Power BI? Variance is calculated in Power BI using DAX (Data Analysis Expressions):

```
-- Sample Variance
```

```
Sales_Variance_Sample = VAR.S(Sales[SalesAmount])
```

```
-- Population Variance
```

```
Sales_Variance_Pop = VAR.P(Sales[SalesAmount])
```

1. Real Business Scenario A real company problem: Suppose 'FoodPanda' wants to evaluate the performance of their delivery partners. They have two delivery men (Rahim and Karim). Their delivery times (in minutes) over the last 5 days are as follows:

Rahim: 30, 32, 28, 31, 29 (Average = 30 minutes) Karim: 15, 45, 20, 50, 20 (Average = 30 minutes)

Step-by-step solution:

1. Mean Analysis: Both have an average delivery time of 30 minutes. Looking only at the average, it seems both are equally efficient.
2. Variance Calculation:
Rahim's Variance: $\text{VAR.S}(30, 32, 28, 31, 29) = 2.5$
Karim's Variance: $\text{VAR.S}(15, 45, 20, 50, 20) = 262.5$
3. Conclusion: Rahim's variance is very low (2.5), meaning he consistently delivers close to 30 minutes. Karim's variance is very high (262.5), meaning sometimes he delivers very quickly, and other times very late. To maintain good customer experience, FoodPanda will consider Rahim more reliable.
4. Interview Perspective
5. What is variance and why is it used? Sample Answer: Variance is a measure of data dispersion that shows how far data points are from their mean. It indicates the consistency of data...

(Consistency) and Risk are understood using these concepts.

1. What is the difference between Sample Variance and Population Variance?
Sample Answer: Population Variance is calculated over the entire dataset (divided by N). Sample Variance is calculated over a smaller portion of the data (Sample) and divided by N-1, which is called Bessel's correction to reduce bias.

2. What is the relationship between Variance and Standard Deviation?

Sample Answer: Standard Deviation is the square root of Variance. The unit of Variance is the square of the original data unit (e.g., square of currency), which is hard to interpret. Therefore, Standard Deviation is used, which has the same unit as the original data.

3. Can Variance ever be negative?

Sample Answer: No, Variance can never be negative because it is calculated by squaring the distance from the mean, and the square of any number is never negative. Its minimum value can be zero (if all data points are the same).

4. If 5 is added to each value in a dataset, how does the Variance change?

Sample Answer: Variance does not change. Because adding 5 to each value increases the mean by 5, the deviation of each data point from the mean remains the same.

1. Beginner Mistakes

What mistakes do beginners commonly make?

2. Confusing Sample and Population: Not understanding when to divide by N and when by N-1.
3. Forgetting the Unit: Forgetting that the unit of Variance is the square of the original data unit (e.g., if data is in 'Taka', Variance is in 'Taka squared').
4. Not understanding the impact of Outliers: Variance is heavily influenced by outliers (extremely large or small values) because it squares the deviations. Beginners often calculate Variance without removing outliers.
5. Drawing conclusions based only on Variance: It is difficult to understand data spread directly from Variance values. Using Standard Deviation is more convenient.
6. Getting negative Variance: Calculation errors leading to negative Variance, which is mathematically impossible.

How to avoid these?

- Confirm whether the dataset is a Sample or Population beforehand.
- Use Standard Deviation instead of Variance in reports or presentations.
- Use Boxplot or Scatter plot to check for outliers during data cleaning.

1. Formula

Population Variance Formula:

$$[\sigma^2 = \frac{\sum (X_i - \mu)^2}{N}]$$

Sample Variance Formula:

$$[s^2 = \frac{\sum (X_i - \bar{x})^2}{n - 1}]$$

Explanation of each part/symbol in the formula:

- (σ^2) (Sigma squared) = Population Variance
- (s^2) = Sample Variance
- $(\sum)$ (Sigma) = Summation
- (X_i) = Each data point
- (μ) (Mu) = Population Mean (mean of the entire dataset)
- $(\bar{x})$ (x-bar) = Sample Mean (mean of the sample data)
- (N) = Total number of data points in Population
- (n) = Total number of data points in Sample
- $(n - 1)$ = Degrees of freedom (Bessel's correction)

1. Practical Exercise

Beginner Exercise (Easy):

1. Sales of a shop for 5 days (in Taka): 100, 120, 110, 130, 90. Calculate the Sample Variance.
2. Ages of 5 students: 20, 21, 20, 22, 21. Calculate the Population Variance.
3. Weights of 5 balls produced by a machine (in grams): 50, 50, 50, 50, 50. What is the Variance? (Answer: 0)

Intermediate Exercise (Moderate difficulty):

1. Prices of stocks of two different companies over the last 10 days are given. Calculate the Variance to prove which company's stock is more risky to invest in.
2. A dataset has a Variance of 144. What is the Standard Deviation? If each data value is multiplied by 10, what will be the new Variance?

Real Business Exercise (Real company problem):

Problem: 'Pathao' is analyzing rider ratings (1-5). In Zone A, riders have an average rating of 4.2 and Variance of 0.5. In Zone B, riders have the same average rating of 4.2 but Variance of 1.8.

Task: As a Data Analyst, explain to management which zone's riders have more consistent performance and which zone requires more training.

10. Career Perspective

Junior Data Analyst: Primarily calculates the variance and standard deviation.
Data Analyst: Analyzes variance to extract business insights. For example
Senior Data Analyst: Uses variance thresholding for feature selection when
Business Analyst: Focuses more on the business impact of variance rather than

Summary and Cheat Sheet Key points summarized:

Variance measures how much data is spread from its mean.
High variance means more diversity or risk in the data; low variance means
Divide by N for population and by N-1 for sample.
The square root of variance is the standard deviation.

Quick Reference Cheat Sheet: Concept Description Excel Formula SQL Function Python (Pandas)

Population Entire =VAR.P() VAR_POP() df.var(ddof=0) Variance dataset's

spread

Sample Sample =VAR.S() VAR_SAMP() df.var() Variance data spread

Standard Square root =STDEV.S() STDDEV_SAMP() df.std() Deviation of variance

Key Rule 1 Variance - - -

is never
negative

Key Rule 2 Adding or - - -

subtracting
a constant
to data does
not change
variance

Key Rule 3 Multiplying - - -

data by a
constant
multiplies
variance by
the square of
that constant

Normal Distribution (স্বাভাবিক বিন্যাস) ()

1. Definition What is it? Normal Distribution is one of the most important probability distributions in statistics. It forms a bell-shaped curve where most data points cluster around the mean, and the further you move away from the mean, the fewer data points there are. It is symmetrical, meaning the left and right sides of the mean look exactly the same.

Explanation in simple Bengali and everyday example: Suppose you measure the heights of adult men in a neighborhood in Dhaka. You will find most people's heights fall between 5 feet 5 inches and 5 feet 8 inches. Very few people are shorter than 5 feet (very short), and very few are taller than 6 feet 2 inches (very tall). If you plot this height data on a graph, it will form a bell-shaped curve. The middle (mean height) will have the highest number of people, and the numbers will decrease towards both ends. This is the Normal Distribution.

1. Why is it important? Why is it used in Data Analytics?

Predictability: Normal Distribution follows a specific pattern (such as the Empirical Rule: 68-95-99.7). This helps us easily understand how common or unusual a data point is.

Hypothesis Testing: Most statistical tests (like t-test, ANOVA) assume that data follows a Normal Distribution. Outlier Detection: Helps identify abnormal values (outliers) in the data.

Importance in Business Decisions:

Risk Management: To measure the probability of stock returns or risks in .
Quality Control: To verify whether product quality in manufacturing is con
Inventory Management: To forecast customer demand and maintain the right :

1. How do companies use it? E-commerce (e.g., Daraz): Analysis of delivery times. Most deliveries occur within 3-4 days (average). Very

Delivery is done in 1 day or 10 days. Daraz uses this distribution to give customers an accurate delivery time commitment.

Banking (e.g., bKash, City Bank): Analyzing customers' credit scores for credit cards or loans. Most people's scores are average. Very few have very bad or very good scores. Based on this, banks decide on loan approvals.

Healthcare (e.g., Square Hospital): Analyzing patients' blood pressure or cholesterol levels. Normal people's blood pressure follows a Normal Distribution. Values outside this range (Outliers) are marked as risky.

Telecom (e.g., Grameenphone): Daily internet usage by customers. Most customers use on average 1-2 GB. Very few use none or up to 10 GB. Based on this, GP designs new data packages.

Marketing (e.g., Unilever): Analyzing customers' response time or click-through rate (CTR) in advertisements. In a campaign, most customers respond within a specific time frame.

Retail (e.g., Shwapno): Number of customers visiting the store daily. Except holidays, the number of customers on normal days follows a normal distribution, which is used for staff scheduling.

Manufacturing (e.g., Walton): Size or weight of refrigerator or air conditioner parts. If a certain part is supposed to weigh 5 grams, most parts will be close to 5 grams. Any deviation beyond this is rejected.

-
1. How does a Data Analyst use it?

How to use it with Excel?

Function: `NORM.DIST(x, mean, standard_dev, cumulative)`

Example: Suppose average sales are 100 and Standard Deviation is 15. You want to know the probability of sales being below 90.

Formula: `=NORM.DIST(90, 100, 15, TRUE)`

How to use it with SQL?

SQL does not have a direct Normal Distribution function, but Mean and Standard Deviation can

be calculated to compute the Z-score.

```
WITH Stats AS (  
    SELECT AVG(sales) as mean_sales, STDDEV(sales) as std_sales  
    FROM daily_sales  
)  
SELECT  
    date,  
    sales,  
    (sales - mean_sales) / std_sales AS z_score  
FROM daily_sales, Stats  
WHERE ABS((sales - mean_sales) / std_sales) > 3; -- Outlier detection
```

How to use it with Python?

```
import numpy as np  
import scipy.stats as stats  
import matplotlib.pyplot as plt  
  
# Generate data  
data = np.random.normal(loc=50, scale=5, size=1000) # Mean=50, Std=5  
  
# Probability Density Function (PDF)  
x = np.linspace(30, 70, 100)  
y = stats.norm.pdf(x, loc=50, scale=5)  
  
plt.plot(x, y)  
plt.title('Normal Distribution')  
plt.show()
```

How to use it with Power BI?

You can create a Normal Distribution Curve using DAX.

Normal_Distribution = VAR MeanValue = AVERAGE('Sales'[Amount]) VAR StdDevValue =
STDEV.P('Sales'[Amount]) VAR X = MAX('Sales'[Amount]) RETURN

```
EXP( - (X - MeanValue)^2 / (2 * StdDevValue^2) ) / (StdDevValue * SQR
```

1. Real Business Scenario

Problem: Foodpanda takes on average 30 minutes to deliver food in a specific area, with a Standard Deviation of 5 minutes. The company wants to know what percentage of deliveries take more than 40 minutes so they can budget for customer compensation for late deliveries.

Solution (Step by step):

1. Understand the data: Mean (μ) = 30, Standard Deviation (σ) = 5, Target (X) = 40.
 2. Calculate Z-score: $Z = (X - \mu) / \sigma = (40 - 30) / 5 = 2$.
 3. Find Probability: A Z-score of 2 means 40 minutes is 2 Standard Deviations above the mean. Using a Z-table or Excel ($=1 - \text{NORM.S.DIST}(2, \text{TRUE})$), the value is approximately 0.0228 or 2.28%.
 4. Conclusion: Only 2.28% of deliveries take more than 40 minutes. Foodpanda will allocate a compensation budget for this 2.28% of orders.
-

1. Interview Perspective

Question 1: What is Normal Distribution and what are its characteristics?

Sample Answer: Normal Distribution is a continuous probability distribution shaped like a bell curve. Its main characteristics are: Mean, Median, and Mode are equal; it is symmetrical; and it follows the 68-95-99.7 rule.

Question 2: What is the Empirical Rule (68-95-99.7 Rule)?

Sample Answer: According to this rule, in a Normal Distribution, approximately 68% of data lies within 1 Standard Deviation of the mean, 95% within 2 Standard Deviations, and 99.7% within 3 Standard Deviations.

Deviation lies within, and 99.7% of the data lies within 3 Standard Deviations.

Question 3: What is a Z-score and why is it used?

Sample Answer: A Z-score is a measure that specifies how many Standard Deviations a data point is away from the mean. It is used to compare data on different scales and to identify outliers.

Question 4: What will you do if your data does not follow a Normal Distribution?

Sample Answer: If the data is not normal, I can use data transformation techniques (such as Log transformation, Square root transformation). Alternatively, I can use non-parametric statistical tests that do not assume normality of the data.

Question 5: What is the relationship between the Central Limit Theorem (CLT) and Normal Distribution?

Sample Answer: The CLT states that if we take sufficiently large samples from any population (regardless of distribution), the distribution of the sample means will be approximately Normal.

1. Beginner Mistakes

Mistake 1: Assuming all data follows a Normal Distribution.

How to avoid: Before analyzing data, check the distribution using a Histogram or Q-Q plot.

Mistake 2: Considering outliers as normal data.

How to avoid: Identify outliers using Z-score or Boxplot and handle them separately.

Mistake 3: Not understanding the difference between Mean and Median.

How to avoid: Use Median instead of Mean for skewed data because Mean does not represent the data accurately if the distribution is not normal.

Mistake 4: Ignoring the importance of Standard Deviation.

How to avoid: Always check Standard Deviation to understand data spread, not just the Mean.

Mistake 5: Expecting Normal Distribution with a small sample size.

How to avoid: Use t-distribution when the sample size is small (e.g., less than 30).

1. Formula

2. Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}$$

- x : Data point (Variable)
- μ (Mu): Population mean
- σ (Sigma): Standard Deviation
- π : Constant (Pi, approximately 3.14159)
- e : Constant (Euler's number, approximately 2.71828)

1. Z-score Formula:

$$Z = \frac{X - \mu}{\sigma}$$

- Z : Z-score
- X : Specific data value
- μ : Mean
- σ : Standard Deviation

1. Practical Exercise

Beginner Exercise:

2. The average score of a class is 70 with a Standard Deviation of 10. What percentage of students scored between 60 and 80? (Hint: Use the Empirical rule).
3. Run the formula `=NORM.DIST(85, 75, 8, TRUE)` in Excel and explain its meaning.
4. Take 5 different height data points and manually calculate their Z-scores.

Intermediate Exercise:

1. Using Python, plot a Normal Distribution graph where Mean=100 and Std=15.
2. Using SQL, retrieve the days from a sales table where sales were more than 2 Standard Deviations above the average.

Real Business Exercise:

Problem: The average number of daily trips per Pathao rider is 15 with a Standard Deviation of 3. If a rider must complete at least 21 trips in a day to earn a bonus, what percentage of riders will receive the bonus? (Solve using Z-score and Probability).

1. Career Perspective

Junior Data Analyst: Primarily uses Excel or Python to check data distribution (by creating Histograms) and performs basic outlier removal.

Data Analyst: Uses SQL and Python to calculate Z-scores on large datasets, verifies data normality for A/B testing, and presents insights on dashboards.

Senior Data Analyst: Develops complex statistical models, designs sampling strategies using the Central Limit Theorem, and performs feature engineering for predictive analysis.

Business Analyst: Uses insights from Normal Distribution to make business decisions. For example, "95% of our customers receive a response within 5-15 minutes, so our SLA (Service Level...

Agreement) should be set to 15 minutes."

Summary and Cheat Sheet

Summary: Normal Distribution is a bell-shaped distribution where most data points are close to the mean. It is used to understand data patterns, detect outliers, and perform various statistical tests.

Cheat Sheet:

- Shape: Bell-shaped, Symmetrical.
- Center: Mean = Median = Mode.
- Empirical Rule:

- $\mu \pm 1\sigma = 68\%$ of data
- $\mu \pm 2\sigma = 95\%$ of data
- $\mu \pm 3\sigma = 99.7\%$ of data
- Z-score: $Z = (X - \mu)/\sigma$ (a measure of distance from the mean).
- Excel Functions: NORM.DIST(), NORM.INV(), STANDARDIZE().
- Python Library: scipy.stats.norm.

Percentiles & Quartiles

1. Definition

What is it? Percentiles and Quartiles are two statistical measures that divide a dataset into equal parts.

- Percentile: Divides a dataset into 100 equal parts. For example, the 90th percentile means 90% of the data values are below or equal to that specific value.
- Quartile: Divides a dataset into 4 equal parts.
 - Q1 (First Quartile / 25th Percentile): The lowest 25% of the data.
 - Q2 (Second Quartile / 50th Percentile / Median): The middle value (50%).
 - Q3 (Third Quartile / 75th Percentile): The start of the top 25% (75%).

Explanation in simple terms with everyday examples: Suppose you took a competitive exam and your result card says you scored in the “95th Percentile.” This does not mean you scored 95 out of 100. It means your score is higher than or equal to 95% of the participants. Regarding quartiles, imagine a class of 20 students arranged by height from shortest to tallest. The first 5 students represent Q1, the next 5 Q2, then Q3, and the last 5 Q4.

1. Why is it important?

Why is it used in Data Analytics?

- Outlier Detection: Quartiles (especially the IQR - Interquartile Range) are used to identify unusually high or low values (outliers) in a dataset.
- Understanding Data Distribution: Helps to understand how data is spread.
- Benchmarking: Used to compare performance standards.

Importance in Business Decisions:

- Customer Segmentation: Dividing customers into groups based on spending (e.g., Top 10% customers).

- Salary Benchmarking: HR departments use the market 75th percentile to set employee salaries.
- Inventory Management: Identifying the top 25% best-selling products to manage stock.

1. How do companies use it?

- E-commerce (e.g., Daraz, Amazon): Create quartiles based on customer spending. Those in Q4 (Top 25%) receive special discounts or premium memberships.
- Banking (e.g., bKash, City Bank): Use income percentiles to set credit card limits. Customers in the 90th percentile income group get higher limits.
- Healthcare (e.g., Square Hospital): Analyze patient waiting times. If the 80th percentile waiting time is 45 minutes, it means 80% of patients wait 45 minutes or less.
- Telecom (e.g., Grameenphone): Segment users based on internet data usage. Heavy users in the Top 5% (95th percentile) get special data packages.
- Marketing: Analyze campaign response rates to identify campaigns performing in the Top 10% (90th percentile).
- Retail (e.g., Shwapno): Compare store sales performance and identify stores above the 75th percentile.
- Manufacturing (e.g., Walton): Quality control by rejecting batches with defect rates above the 95th percentile.

1. How does a Data Analyst use it?

How to use it in Excel?

- Percentile: =PERCENTILE.INC(A2:A100, 0.9) (to find the 90th percentile)
- Quartile: =QUARTILE.INC(A2:A100, 1) (to find Q1), =QUARTILE.INC(A2:A100, 3) (to find Q3)

How to use it in SQL?

```
-- Query to find the 90th Percentile
SELECT
    PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY sales_amount) AS percenti
FROM sales_table;
```



```

FROM sales_data;

-- Query to find Quartiles
SELECT
    NTILE(4) OVER (ORDER BY sales_amount) AS quartile,
    sales_amount
FROM sales_data;

```

How to use it with Python?

```

import pandas as pd
import numpy as np

# Create DataFrame
data = {'Sales': [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]}
df = pd.DataFrame(data)

# Calculate Percentile
p90 = np.percentile(df['Sales'], 90)
print(f"90th Percentile: {p90}")

# Calculate Quartiles
q1 = df['Sales'].quantile(0.25)
q3 = df['Sales'].quantile(0.75)
iqr = q3 - q1
print(f"Q1: {q1}, Q3: {q3}, IQR: {iqr}")

```

How to use it in Power BI? Using DAX Measure:

- 90th Percentile Sales_90th_Percentile = PERCENTILE.INC(Sales[Amount], 0.90)
 - Q1 (First Quartile) Sales_Q1 = PERCENTILE.INC(Sales[Amount], 0.25)
5. Real Business Scenario A real problem faced by a company: Suppose Foodpanda is facing issues with their delivery time. Many customers are complaining that the food is arriving very late. Management wants to understand the current status of delivery time and how it can be improved. Solution (Step by step):

1. Data Collection: Collected all delivery time data from the last month.

2. Average vs Percentile: Looking only at the average delivery time does not show the full picture because a few very high or very low delivery times can skew the average.
3. Percentile Calculation: The Data Analyst calculated the 50th (Median), 90th, and 95th percentile delivery times.

50th Percentile (Median): 25 minutes (i.e., 50% of deliveries took 25 minutes or less)
90th Percentile: 45 minutes (i.e., 90% of deliveries took 45 minutes or less)
95th Percentile: 60 minutes (i.e., 5% of deliveries took more than 60 minutes)

4. Decision Making: Management saw that 5% of customers are waiting more than 1 hour, which is harmful to the brand image. They decided to reduce the 95th percentile delivery time from 60 minutes to 40 minutes.
5. Action: More riders were assigned to areas with longer delivery times, and discussions were held with restaurants to reduce food preparation time.
6. Interview Perspective
7. What is the difference between Percentile and Percentage? Sample Answer: Percentage is a specific amount out of 100 (e.g., scoring 80 out of 100 marks means 80%). On the other hand, Percentile is a comparative position. The 80th percentile means you performed better than 80% of people.
8. How do Quartiles help in detecting Outliers? Sample Answer: We use the Interquartile Range (IQR) to detect outliers. $IQR = Q3 - Q1$. If any value is smaller than $Q1 - 1.5 * IQR$ or larger than $Q3 + 1.5 * IQR$, it is considered an outlier.
9. When should Median (50th Percentile) be used instead of Mean? Sample Answer: When the dataset contains many outliers or is skewed, the mean does not represent the data accurately. In such cases, using the median or 50th percentile is more appropriate.
10. What does the NTILE function do in SQL? Sample Answer: The NTILE function divides a dataset into a specified number of equal parts. For example, NTILE(4) divides the data into 4 quartiles.
11. What does the 99th Percentile mean? Give a real example. Sample Answer: The 99th percentile means 99% of the data values fall below that point. For example, if a website's response time at the 99th percentile is 2 seconds, it means 99% of users experience the website loading in 2 seconds or less.
12. Beginner Mistakes Common mistakes made by beginners:

1. Confusing Percentage and Percentile: Many think the 90th percentile means scoring 90%.
2. Not checking for Outliers: Starting data analysis without checking for outliers using Quartiles/IQR.
3. Using Mean in all cases: Using mean instead of median without understanding the data distribution.
4. Forgetting Quartile 0 and 4: Overlooking Q0 (Minimum) and Q4 (Maximum), which help understand the data range.
5. Using wrong functions: Not understanding the difference between INC (Inclusive) and EXC (Exclusive) functions in Excel or Power BI.

How to avoid these?

Keep the basic concepts clear.

Always create a Box Plot before data analysis to observe the data distribution and identify outliers.

Read the documentation of tools (Excel/SQL/Python) to understand the correct usage of functions.

1. Formula

Percentile Formula:

$$[P_k = \frac{k}{100} \times (n + 1)]$$

- (P_k) = Position of the k-th Percentile
- (k) = The desired percentile (e.g., 90)
- (n) = Total number of data points in the dataset

Quartile Formula:

$$[Q1 \text{ (First Quartile)} = \frac{1}{4} \times (n + 1)^{\text{th}} \text{ value}]$$

$$[Q2 \text{ (Median)} = \frac{2}{4} \times (n + 1)^{\text{th}} \text{ value}]$$

$$[Q3 \text{ (Third Quartile)} = \frac{3}{4} \times (n + 1)^{\text{th}} \text{ value}]$$

$$[IQR \text{ (Interquartile Range)} = Q3 - Q1]$$

1. Practical Exercise

Beginner Exercise (Easy):

2. Data: [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]. Find the Median (50th Percentile) of this data.

3. Data: [5, 15, 25, 35, 45, 55, 65]. Find Q1 and Q3 of this data.

4. Data: [12, 24, 36, 48, 60]. Find the 80th Percentile of this data.

Intermediate Exercise (Moderate difficulty):

1. Weights (kg) of students in a class: [45, 50, 52, 55, 58, 60, 62, 65, 70, 120]. Identify outliers using IQR.
2. Write an SQL query to find the 75th Percentile of salary from an employees table.

Real Business Exercise (Real company problem):

Problem: You are a Data Analyst at 'Pathao'. You have ride fare data for the past week.

Management wants to know the minimum fare among the top 10% highest fares.

Task: Use Python or Excel to calculate the 90th Percentile fare and write a brief report for management.

1. Career Perspective

- Junior Data Analyst: Primarily calculates Percentiles and Quartiles for various reports using Excel or SQL.
- Data Analyst: Uses Python/R to remove outliers from datasets and creates Box Plots to visualize data distribution.
- Senior Data Analyst: Builds predictive models using percentile data and supports strategic decision-making in the company (e.g., pricing strategy).
- Business Analyst: Analyzes percentile/quartile reports provided by Data Analysts to extract business insights and presents them to stakeholders.

Summary and Cheat Sheet

Key Points:

- Percentiles divide data into 100 parts, Quartiles divide data into 4 parts.
- ($Q1 = 25^{\text{th}}$) Percentile, ($Q2 = 50^{\text{th}}$) Percentile (Median), ($Q3 = 75^{\text{th}}$) Percentile.
- IQR ($Q3 - Q1$) is crucial for identifying outliers.
- Widely used in business for customer segmentation, performance benchmarking, and risk management.

Quick Reference Cheat Sheet:

Concept	Description	Excel Formula	Python (Pandas)
25th Percentile	Q1 (First Quartile)	=QUARTILE.INC(range, 1)	df['col'].quantile(0.25)
50th Percentile	Q2 (Median)	=MEDIAN(range)	df['col'].median()
75th Percentile	Q3 (Third Quartile)	=QUARTILE.INC(range, 3)	df['col'].quantile(0.75)
IQR	Q3 - Q1	=QUARTILE(range,3) - QUARTILE(range,1)	q3 - q1
Outlier Lower Limit	Q1 - (1.5 * IQR)	-	q1 - 1.5 * iqr
Outlier Upper Limit	Q3 + (1.5 * IQR)	-	q3 + 1.5 * iqr